# What carries the gains of debiased graph recommendation? A controlled study on four benchmarks with unbiased test data

Anonymous

**Abstract**

Under one controlled protocol, the gains reported for graph exposure-bias corrections depend strongly on evaluation choices, and third-party corrections add little over an equally tuned backbone. We re-evaluate these corrections on Coat, Yahoo! R3, KuaiRand-Pure and KuaiRec. Uniform data serve only for validation and testing, except Coat's shipped propensities, which are checked for the prior rows only. Configurations are selected per split on validation data, and leading methods are retrained with three seeds. Four findings follow. Selecting on test data inflates Coat scores by up to 0.030 nDCG@5. Third-party debiasers improve LightGCN by point-estimate gains of at most $4.10 \times 10^{-3}$ nDCG on Coat, Yahoo! R3 and KuaiRand-Pure, and none survives a correction for overlapping splits there. An additive prior fitted on the exposed log gives the largest point-estimate gains over LightGCN, $31.98 \times 10^{-3}$ on Coat and $21.22 \times 10^{-3}$ on KuaiRec. The Coat gain is not robust to that correction, the KuaiRec gain is, and the mechanism is not identified. Making the multi-hop propagation exactly unbiased does not improve ranking on the three real logs where it was run. The scope is four public benchmarks, offline evaluation and implicit-feedback learners. We propose no new estimator.

*Keywords:* recommender systems, exposure bias, graph collaborative filtering, propensity, doubly robust

## 1. Introduction

Under one controlled protocol with unbiased test data, the gains reported for exposure-bias corrections of graph recommenders depend strongly on eval-

uation choices, and third-party corrections add little over an equally tuned backbone.

Graph-based recommenders represent users and items as nodes and score a user-item pair by propagating embeddings over the user-item graph. LightGCN (He et al., 2020) is a standard backbone of this family, and it is usually trained on implicit feedback logs. In such a log an interaction is recorded only when the item was shown and the user responded. The logged graph therefore reflects the exposure policy as well as user taste (Marlin and Zemel, 2009; Chen et al., 2023), and a model fitted to the log treats every absence as informative. The pairwise loss that trains LightGCN, Bayesian personalised ranking (Rendle et al., 2009), draws its negatives uniformly from the items the user has no logged interaction with. Under exposure bias most of these items were never shown. Corrections act on the loss (Schnabel et al., 2016; Wang et al., 2019b; Guo et al., 2021), on the latent exposure or on the likelihood weights (Liang et al., 2016; Saito et al., 2020), and, for graph models, on aggregation or adjacency (Kim et al., 2022; Zhao et al., 2022; Islam et al., 2026b).

This paper starts from the two nuisance functions that a DR estimator already fits: an exposure propensity $p_{ui}$ and an imputed preference $\hat{Y}_{ui}$. Under an independence assumption between exposure and preference given covariates, Bayes' rule gives the probability that a pair absent from the log is a positive,

$$Q_{ui} = \Pr\big(Y_{ui} = 1 \mid O_{ui}Y_{ui} = 0,\, x_{ui}\big) = \frac{\hat{Y}_{ui}\,(1 - p_{ui})}{1 - p_{ui}\,\hat{Y}_{ui}}. \tag{1}$$

The candidate models use $Q$ in two places inside an otherwise unchanged LightGCN. Each sampled BPR negative is down-weighted by $1 - Q_{uj}$, and the standardised logit of the imputation is added to the score as a fixed offset whose weight is chosen on validation data. Both nuisance functions are fitted on the biased log before training. The one exception to the rule that training uses no uniformly collected data is Coat, whose shipped propensities were estimated by the dataset authors from a uniform sample. We call the full model QP-LightGCN and evaluate it with its variants as one family.

The gains reported for these corrections are hard to compare, for three reasons. First, the studies use uniformly collected data in different ways: some train on it, some select configurations on it, and some use it only for testing (Section 2). Second, tuning budgets differ, so a method given a larger

search space can obtain a higher test score through selection alone. Third, single runs are common, and the ordering of close methods can depend on the random seed. Our own runs illustrate the third point. In the main run, QP-LightGCN is above iALS on Coat by $+20.02\times10^{-3}$ [19.74], which is nominally significant after Holm correction over the eight baseline comparisons ($p_{\text{Holm}} = 0.047$). When the selected configurations are retrained with seeds 1 to 3, the difference is $+3.39 \times 10^{-3}$ [20.70], with $p_{\text{Holm}} = 0.75$ over the two baseline comparisons of that check. These seed comparisons are descriptive, because the two baselines of each dataset were chosen on test data.

The study asks four research questions, which Table 1 answers and locates.

Table 1: Research questions, short answers and the sections that answer them.

| Question | Short answer | Section |
|---|---|---|
| RQ1: How much do evaluation choices change the conclusions? | A lot. Selection on the test split inflates scores and can reverse an order. Many pairwise signs depend on the search budget, the seed or the metric. | Section 5.1 |
| RQ2: How much do graph-specific corrections add over an equally tuned backbone? | Little, on Coat, Yahoo! R3 and KuaiRand-Pure. On KuaiRec the gains are larger, and a pointwise loss without debiasing gives a larger difference of means. | Section 5.2 |
| RQ3: Which ingredient carries the gains that remain? | An additive prior fitted once on the exposed log. Its inverse-propensity weights are not what matters in our runs, and the mechanism is not identified. | Section 5.3 |
| RQ4: Does exact unbiasedness of the propagation matter for ranking? | Not on the three real logs on which it was run. | Section 5.4 |

The paper makes four contributions. None of them is a new estimator.

1. *Protocol and its sensitivity (RQ1).* Four public benchmarks with unbiased test data, uniform data used only for validation and testing, per-split selection on validation data, per-user paired tests with Holm correction, and retraining with seeds. Selecting on the test split inflates Coat scores by up to 0.030 nDCG@5.
2. *Third-party debiasers add little over an equally tuned backbone (RQ2).* On Coat, Yahoo! R3 and KuaiRand-Pure the largest point-estimate gain of a third-party debiaser over LightGCN is $+4.10 \times 10^{-3}$ nDCG,

and none of these gains survives a correction for overlapping splits on those datasets.

3. *Prior anatomy (RQ3).* The gains over the backbone that the additive prior provides are examined with controls. Removing its inverse-propensity weights does not reduce them, a shuffled item term removes the KuaiRec gain, and popularity does not reproduce it. The mechanism is not identified.
4. *A negative result on walk correction (RQ4).* The walk correction of DRUP removes the repeated-edge bias of multi-hop DR propagation. It does not improve ranking on Coat, Yahoo! R3 or KuaiRec.

The QP-LightGCN family and its refinements are members of the evaluated family (Appendix B), not contributions in their own right. We do not claim that debiasing is useless in general.

The scope is four public benchmarks, offline evaluation, top-$K$ nDCG at the cut-off used for each dataset (5 for Coat and Yahoo! R3, 10 for KuaiRand-Pure, 20 for KuaiRec), and implicit-feedback learners trained on the biased log. The walk-correction claim concerns Coat, Yahoo! R3 and KuaiRec, the three real logs on which DR and DRUP were run. Coat has 225 evaluated users, so its power is low. For Coat we report the differences that the data exclude, and we do not report equality. We do not claim that AdvInfoNCE, NAVIP, r-AdjNorm, SimGCL or MACR fail in general. Our implementations, tuned with the stated budgets under this protocol, do not beat the backbone by more than the amounts reported. We do not give a mechanism for why the prior helps beyond the controls of Section 5.3, and the reading that the prior acts as a debiased item-quality score is a hypothesis.

## 2. Related work

### *2.1. Graph collaborative filtering and graph debiasing*

LightGCN (He et al., 2020) is the backbone used in our experiments. Other graph convolutional recommenders include NGCF (Wang et al., 2019a) and UltraGCN (Mao et al., 2021), and contrastive graph variants include SGL (Wu et al., 2021) and SimGCL (Yu et al., 2022). Linear and closed-form models include EASE (Steck, 2019), GF-CF (Shen et al., 2021) and BSPM (Choi et al., 2023). Weighted matrix factorisation for implicit feedback, iALS (Hu et al., 2008), is a standard baseline, and Rendle et al. (2022) revisit its tuning on item recommendation benchmarks. Wu et al. (2023) survey graph

neural recommenders. These models fit or propagate over the logged graph and do not model which pairs were exposed.

Graph models have also been debiased. NAVIP (Kim et al., 2022) weights neighbour aggregation by inverse propensities. r-AdjNorm (Zhao et al., 2022) tunes the strength of adjacency normalisation to trade accuracy against novelty, and Chen et al. (2024) study how graph convolutions can amplify popularity bias. APDA (Zhou et al., 2023) and CAGED (Que et al., 2025) learn aggregation weights, and DPAA (Islam et al., 2026a), an arXiv preprint, reweights message passing. AdvDrop (Zhang et al., 2024) learns an adversarial edge dropout on the interaction graph and reports IPS and DR baselines on Coat, Yahoo! R3 and KuaiRec. PPD (Islam et al., 2026b) removes a popularity direction from pre-trained LightGCN embeddings without retraining, and is evaluated on KuaiRec, Coat and Yahoo! R3. DR-GNN (Wang et al., 2024) applies distributionally robust optimisation over graph neighbourhoods; its "DR" stands for distributionally robust, not doubly robust. The popularity-debiasing methods MACR (Wei et al., 2021) and PDA (Zhang et al., 2021) are used as baselines in our study. Most of these methods act on aggregation, adjacency, embeddings or a robust objective, rather than on the negatives of the pairwise loss.

### *2.2. Propensity, doubly robust and negative-weighting methods*

Ratings and implicit feedback are missing not at random (Marlin and Zemel, 2009), and surveys organise the resulting biases and the causal methods that address them (Chen et al., 2023; Gao et al., 2024). Propensity methods re-weight the loss by inverse propensities or self-normalised estimators (Schnabel et al., 2016; Swaminathan and Joachims, 2015; Saito et al., 2020). Steck (2010) showed that missing ratings carry information, proposed test measures that can be estimated without bias when ratings are missing not at random, and derived training objectives that account for the missing entries. Doubly robust learning adds an imputation model to the inverse-propensity estimator (Wang et al., 2019b), and later doubly robust variants address related debiasing problems (Guo et al., 2021; Dai et al., 2022; Li et al., 2023). Information-theoretic counterfactual learning from missing-not-at-random feedback is studied by Wang et al. (2020).

Some debiasing methods use a small uniform sample for training. AutoDebias (Chen et al., 2021) learns sample weights and an imputation function by bi-level meta-learning on a small uniform sample. CausE (Bonner and Vasile, 2018) embeds a small randomised sample alongside the logged

one, and KDCRec (Liu et al., 2020) distils from a model fitted on uniform data into a model trained on the biased log.

The methods closest to our negative weighting are those for exposure and false negatives. ExpoMF (Liang et al., 2016) infers an exposure posterior for unclicked items inside a matrix factorisation model. Rel-MF (Saito et al., 2020) re-weights observations by propensity, and unbiased pairwise learning (Saito, 2020) applies the same idea to pairwise objectives. SRNS (Ding et al., 2020) identifies likely false negatives during sampling, and debiased contrastive learning (Chuang et al., 2020) corrects negatives that are in fact positives. In contrastive collaborative filtering, the BC loss (Zhang et al., 2022) adds a margin that depends on the popularity-bias degree of each interaction, and AdvInfoNCE (Zhang et al., 2023) assigns each negative an adversarially learned hardness weight. Correct-and-Weight (Yin et al., 2026), an arXiv preprint, weights negatives against false negatives and is tested with graph backbones on random splits. Exposure-aware negative sampling for graph recommenders uses in-community popularity as a proxy for exposure (Chen et al., 2026), also an arXiv preprint. The original formulations of ExpoMF, SRNS, debiased contrastive learning and AdvInfoNCE do not combine a propensity with an imputation of the kind a doubly robust estimator provides. IPS-weighted BPR on LightGCN has been studied with simulated propensities (Raja and Vats, 2025).

The offset used by our candidate models is a fixed, standardised imputation logit added to the score. Adding a fixed offset is a familiar device. In long-tail classification, logit adjustment adds a class prior to the logits (Menon et al., 2021). The offset used here differs in one respect: it is a debiased estimate of item-level preference, obtained from an inverse-propensity-weighted imputation, and not a class prior.

### *2.3. Evaluation and reproducibility*

Several studies have shown that comparisons between recommenders depend on the protocol. Ferrari Dacrema et al. (2019) re-ran recent neural recommenders against simple baselines and found that several did not outperform well-tuned versions of those baselines. Rendle et al. (2019) argue that baselines in published comparisons are often under-tuned, so that a reported gain depends on the effort spent on each method. Rendle et al. (2022) show that iALS, when tuned carefully, is competitive on common item recommendation benchmarks. Anelli et al. (2023) question the reported advances of graph collaborative filtering.

Several parts of the protocol can change a conclusion. Sampled and exact metrics can disagree (Krichene and Rendle, 2020). Nadeau and Bengio (2003) propose a corrected resampled t-test for the generalisation error, for resamples that overlap. Reporting the expected validation performance as a function of the number of configurations tried makes the tuning budget visible (Dodge et al., 2019). Comparisons over several methods and datasets also need a multiple-testing correction, for which we use the step-down procedure of Holm (1979).

### *2.4. Gap and positioning*

The contribution is evidence rather than a new method: the graph debiasers and the candidate models of Section 4 are evaluated under one protocol. Table E.20 shows that related studies differ in how they use uniformly collected data. Some train on it (AutoDebias, CausE, ConAlign), some select configurations on part of it (CDR, PPD, ConAlign), and some keep it out of both (DCE, BISER).

Table E.20 in Appendix E lists each study's protocol.

A literature search in October 2026 over general web sources, which cannot be exhaustive, found no graph recommender that weights sampled negatives by a posterior built from a propensity and an imputation, none that adds a debiased imputation to a graph model's score, and no graph study that uses KuaiRand-Pure. Coat, Yahoo! R3 and KuaiRec are already used for graph debiasing by AdvDrop, PPD and DPAA, so we do not claim that the posterior weighting or the imputation offset is new. The claim is narrower: no study we found evaluates graph debiasers on all four benchmarks under one protocol, with uniform data excluded from training, and analyses how the budget and the selection rule change the conclusions.

## 3. Protocol

The protocol rests on one rule. Uniformly collected data are read only to choose among trained models and to measure them. They never enter a training objective, a nuisance model that a trained model uses, or the stopping of training. This section states the controls that enforce the rule, how the splits, the selection and the seeds are built, how the tests and sensitivity margins are defined, and how the stored results were reconciled before the analysis. The methods to which the protocol is applied are listed in Section 4.

*3.1. Where each data source may enter*

Table 2 lists the stages of the pipeline and the data each stage may read. "Log" is the biased training log, which holds the exposure matrix and the outcomes of the exposed pairs. "Validation" and "test" are the parts of the uniformly collected data that a split assigns to each role.

Table 2: Permitted uses of each data source. The Coat row for the propensity is the one exception to the rule, and it is discussed in the text.

| Stage | Log | Validation | Test |
|---|---|---|---|
| Exposure propensity $\hat{p}$ (Coat: shipped, see text) | yes | no | no |
| Additive imputation $\hat{Y}$, prior $z$, weights $Q$ | yes | no | no |
| Cross-fitted nuisances of the DRGP prior | yes | no | no |
| Training of every model | yes | no | no |
| Choice of configuration and epoch within a split | no | yes | no |
| Early stopping within a split | no | yes | no |
| Reported test score of the chosen model | no | no | yes |

The additive imputation is fitted on the log without cross-fitting, so the same log supplies $\hat{Y}$ and the training data. The DRGP prior of Appendix B.4 is cross-fitted over ten folds before its coefficients are fitted. Test scores are computed after every epoch and stored, but no choice reads them. The sensitivity analysis of the results reads them only to measure what selection on test data would change, and it does not report such a selection as a method.

*The Coat exception..* The shipped Coat propensities were estimated by the dataset authors from a uniformly collected sample (Schnabel et al., 2016). That sample is also the source of our validation and test data, so uniform data reach a training input on Coat, indirectly. Every propensity-based row on Coat uses these propensities: DR-LightGCN, NAVIP, DR-JL, MRDR, DR adjacency, DRUP and the prior rows. To check the effect on the prior rows, we refitted them with propensities estimated from the log by the logistic model of Section 4.3. Table C.16 gives the results. Relative to their no-prior backbones, the prior gains are $+19.44 \times 10^{-3}$ [18.66] for iALS-p and $+33.62 \times 10^{-3}$ [16.72] for LightGCN-p with estimated propensities. With the shipped propensities they are $+16.82 \times 10^{-3}$ [18.04] and $+31.98 \times 10^{-3}$ [20.65]. The paired differences between the two propensity inputs are $+2.63 \times 10^{-3}$ [5.27] for iALS-p and $+1.64 \times 10^{-3}$ [10.30] for LightGCN-p, with $p_{\text{Holm}} = 1.000$

in both cases. The values in brackets are 95% half-widths. The stored no-prior runs of iALS and LightGCN serve as the reference for both settings, because neither reads the propensities in its loss. The check changes only the propensity input of the prior rows. For LightGCN-p the gain with estimated propensities is not removed. For iALS-p the estimated-propensity gain is not significant ($p_{\text{Holm}} = 0.33$), so the statement that the gain remains is a point estimate only. The check does not test the assumption that exposure is independent of the outcome given the covariates, which the closed-form posterior of the method requires. It also does not cover the other propensity-based rows of Coat: DR-LightGCN, NAVIP, DR-JL, MRDR, DR, DRUP and the QP-LightGCN family were not refitted, so their dependence on the shipped propensities is not checked.

### *3.2. Splits*

On Coat, the split is made within each user: 30% of each user's candidate items go to validation and the rest go to test. On Yahoo! R3, KuaiRand-Pure and KuaiRec, the split is made over users: 30% of the test users go to validation, and no user is in both parts of a split. Each split is a function of its seed. Coat uses ten splits (seeds 0 to 9) and the other datasets use five (seeds 0 to 4). On Yahoo! R3, a fixed random 5% of the users (270) is excluded from validation and testing.

A user is a test user of a split only if the user has a test entry in that split. A user can fall in the validation part of every split, so the number of test users in the files of the final method is 225, 2,310, 8,321 and 1,410 on Coat, Yahoo! R3, KuaiRand-Pure and KuaiRec. These are slightly below the evaluated-user counts of Table 3, and equal to them on Coat.

Candidates are ranked for each user within the part being scored, and users with no positive candidate in that part are skipped. We rank all candidates and do not sample negatives, because sampled metrics do not agree with the full ranking metric (Krichene and Rendle, 2020). The metric is nDCG@5 on Coat and Yahoo! R3, nDCG@10 on KuaiRand and nDCG@20 on KuaiRec. Tables report the mean over splits and the standard deviation over splits with ddof = 0. The ensemble stage of Appendix B.3 asserts, for every split it reads, that validation and test users are disjoint (on Yahoo! R3, KuaiRand and KuaiRec) or that validation and test items are disjoint for each user (on Coat).

*3.3. Training and per-split selection*

Each configuration is trained once on the log, and the same trained model is scored on every split. The splits therefore share training and differ only in selection. After each epoch, the model is scored on the validation and test parts of every split. Each split keeps the (configuration, epoch) pair with the highest validation nDCG, and the first maximum wins a tie. A split stops when its validation score has not improved for the patience period: five epochs on Coat and Yahoo! R3, and three epochs on KuaiRand and KuaiRec. The shared run ends when every split has stopped or the epoch limit is reached. Once a split has stopped, its selection is fixed. Later epochs are recorded but cannot be chosen.

The gradient-trained models use epoch limits of 100 (Coat), 60 (Yahoo! R3), 40 (KuaiRand) and 30 (KuaiRec). Batch sizes are 256, 1,024, 8,192 and 32,768. Within a dataset, all methods share the epoch limit, batch size and patience, except AdvInfoNCE on KuaiRec, which uses batch size 4,096. On KuaiRec, each training epoch is capped at 2,000,000 pairs.

Three further quantities are computed from the same per-epoch records for the sensitivity analysis. Each is defined on the rows that the stopping rule of its split allows to be chosen.

- The per-split test oracle is the largest test value of a split over all configurations and epochs. It chooses on test data.
- The configuration-level oracle is the configuration whose mean test value, at its own validation-selected epoch, is largest. It also chooses on test data.
- The expected maximum of the validation-selected test value over $k$ configurations is computed as a function of $k$ (Dodge et al., 2019), and the share of 1,000 random subsets of size $k_{\min}$ in which the sign of a primary comparison flips is reported. Here $k_{\min}$ is the smaller of the two configuration counts of the comparison.

The first two are diagnostics. They are never reported as the result of a method.

The reconciliation of the stored results, which checks that every reported selection can be recomputed from its per-epoch records, is described in Appendix E.4.

*3.4. Seeds*

Each split's selected configuration is retrained with seeds 1, 2 and 3. Each split re-selects its epoch on its own validation part, and the configuration is not changed. The retrained final method is compared with the two baselines that had the highest main-run test means in each dataset: iALS and DR-LightGCN on Coat, r-AdjNorm and LightGCN on Yahoo! R3 and KuaiRand, and DR-JL and MRDR on KuaiRec. The baselines are retrained in the same way. Because these two baselines were chosen on test data, the seed comparisons are descriptive, and they are reported without a significance claim beyond the Holm-adjusted values given in the text; the Holm correction runs over the two tests of each dataset. DRUP is deterministic and is not retrained. The seed-0 values in the tables are the main-run values.

Seeds 1 to 3 were not run for the Coat prior rows (iALS-p and LightGCN-p). Their Coat values come from seed 0 only. The result files also contain re-runs of split 0 with seeds 100 to 102, which are stored separately. These runs are not used in any table.

*3.5. Statistical tests*

Differences are computed per user. For each user, the test score is averaged over the splits in which the user is a test user. For a comparison $A - B$, let $d_u$ be the per-user difference for user $u$. The mean $\bar{d}$ over users is the reported difference, and a paired $t$ test over users gives the two-sided $p$-value and a 95% half-width. Differences are reported as $\times 10^{-3}$ with the half-width in brackets. The reported difference is not the difference of the two table means. The two can have opposite signs. For example, on Coat the difference of the table means of DR-LightGCN and LightGCN is $+4.1 \times 10^{-3}$, while the mean per-user difference is $-0.65 \times 10^{-3}$ [30.16]. The per-user difference is the statistic used throughout.

A difference is called significant only when its Holm-adjusted $p$ is below 0.05. The families are as follows.

**Primary family, per dataset.** Four comparisons, P1 to P4, are defined in code from the stored results. P1 and P4 choose their third-party debiaser and trained comparator, respectively, by mean validation score, and never by test score.

- P1: the third-party debiaser with the highest mean validation score on the dataset, against LightGCN.

- P2: each simple prior against its backbone, that is iALS-p against iALS and LightGCN-p against LightGCN, where both exist on the dataset.
- P3: the best refinement against the simple prior of the same learner. On Coat this gives two comparisons, the QP-LightGCN family against LightGCN-p and DR-iALS against iALS-p. On KuaiRec it gives the QP-LightGCN family against LightGCN-p.
- P4: DRUP against the trained model with the highest mean validation score. This is computed on KuaiRec only.

The primary families contain five comparisons on Coat and KuaiRec and two on Yahoo! R3 and KuaiRand.

**Walk-correction family.** DRUP against DR adjacency, one comparison per dataset.

**Refinement families.** Up to three families per dataset. The first holds each refinement against the simple prior of the same learner. The second, on Coat only, holds DR-iALS against iALS-p. The third holds the shuffled and popularity controls against LightGCN, on the datasets where those controls were run.

**Backbone family.** Every method against LightGCN, reported in Table A.9 to Table A.12 in the appendix.

The Nadeau and Bengio correction (Nadeau and Bengio, 2003) is applied as a second test to the split means. With $J$ splits and split-level differences $e_j$, the variance is $\hat{\sigma}^2 = s_e^2\,(1/J + n_2/n_1)$, where $s_e^2$ is the sample variance of the $e_j$. The statistic $\bar{e}/\hat{\sigma}$ is compared with a $t$ distribution with $J-1$ degrees of freedom, and Holm adjustment is applied to these second $p$-values within the same families. In the split design, the selection part is $n_1$ and holds 30% of the users (or of each user's candidates), and the test part is $n_2$ and holds the remaining 70%. The code uses $n_2/n_1 = (1-\rho)/\rho = 0.7/0.3 = 7/3$ with $\rho = 0.3$, which is the standard ratio with the selection part in the role of the training part. The standard derivation does not apply directly, because the models are trained once on the log and only the selection and test parts vary. With $J = 5$ the variance factor is $1/5 + 7/3 \approx 2.53$. Reading the same 70/30 proportions the other way (test as $n_1$) would give $n_2/n_1 = 3/7$ and a factor of $1/5 + 3/7 \approx 0.63$. The larger factor is used, so the correction

is conservative relative to that reading. For Coat with $J = 10$ the factor is $1/10 + 7/3 \approx 2.43$ (against 0.53). We report the corrected values beside the standard paired test and do not treat either as exact.

Equivalence is assessed with two one-sided tests (Schuirmann, 1987) at margin $\delta$, each at level 0.05. The two null hypotheses are $\bar{d} \leq -\delta$ and $\bar{d} \geq \delta$. The reported TOST $p$-value is the larger of the two one-sided $p$-values. The margins are $\pm 0.010$ nDCG@5 on Coat and Yahoo! R3, and $\pm 0.003$ nDCG on KuaiRand (nDCG@10) and KuaiRec (nDCG@20). They are sensitivity thresholds, chosen before the analysis and not derived from the literature, and TOST is presented as a sensitivity analysis of these thresholds. On Yahoo! R3, KuaiRand and KuaiRec, a TOST result below 0.05 is stated as equivalence within the margin. On Coat, the same result is stated only as the exclusion of differences larger than the margin in absolute value, and equivalence is not claimed. A result inside a margin is never stated as equality.

The minimum detectable effect at 80% power and two-sided level 0.05 is $\mathrm{MDE} = (t_{0.975,\,n-1} + t_{0.80,\,n-1})\, s_d/\sqrt{n}$, where $n$ is the number of users, $s_d$ is the standard deviation of the per-user differences, and $t_{q,\nu}$ is the $q$ quantile of a $t$ distribution with $\nu$ degrees of freedom. The MDE uses only the user-level standard deviation and ignores the dependence between splits, so it is optimistic. For the primary comparisons in the analysis, the MDE is 11.8 to $43.1 \times 10^{-3}$ on Coat, 4.7 to $5.6 \times 10^{-3}$ on Yahoo! R3, 1.5 to $1.9 \times 10^{-3}$ on KuaiRand and 0.7 to $4.9 \times 10^{-3}$ on KuaiRec. The Coat values are large because the Coat analysis has 225 evaluated users, and an MDE of this size bounds what the Coat data can show.

Each user contributes one difference per comparison, which is the mean over the splits in which the user is a test user. The splits share users and candidates, so the paired $t$ test over overlapping splits can be anti-conservative (Nadeau and Bengio, 2003). For this reason the Nadeau and Bengio column is reported beside it for the primary comparisons. The tests treat the training log, the fitted nuisances and the trained models as fixed.

## 4. Methods evaluated

Every method is run by us on the splits and the data preparation of Section 3. No number in this study is taken from the literature. Configuration counts are the numbers stored in the result files. Each count is the product of the search-space values listed below, and those values are the arguments

of the run scripts of the repository. Table 4 lists the counts of every method in the main comparison.

Four refinements of the additive prior were developed by us as candidate methods. They are the Q-weighted negatives, the QP-LightGCN family, the ensemble with DRUP and the leave-edge-out DR prior (DRGP). Appendix B describes them and gives their configuration counts (Table B.13). The members that serve as controls are labelled as controls there, and none of the refinements is claimed as a contribution.

### *4.1. Datasets*

We use four public benchmarks. Each pairs a biased log with a test set collected without the selection of the log, so that a debiasing method can be measured against unbiased outcomes. Table 3 gives the sizes, which were checked against the data loaders of the repository.

Table 3: Datasets. "Log": the biased training data, with its density. "Test": the unbiased data. "Candidates": the unbiased test items per user that are ranked, which are the items never logged for that user. "Evaluated users": users with at least one positive candidate in the full test set. The analysis uses 225, 2,310, 8,321 and 1,410 users (Coat, Yahoo! R3, KuaiRand-Pure, KuaiRec), which differ from the evaluated users of 225, 2,315, 8,347 and 1,411 as explained in Section 3.2.

| Dataset | Users / items | Log (density) | $O$ means | Test | Candidates | Evaluated users |
|---|---|---|---|---|---|---|
| Coat | 290 / 300 | 6,960 (8.0%) | Self-rated | 16 random ratings per user | 14.7 | 225 |
| Yahoo! R3 | 15,400 / 1,000 | 311,704 (2.0%) | Self-rated | 10 random songs, 5,400 users | 10 | 2,315 |
| KuaiRand | 10,000 / 7,583 | 511,863 (0.68%) | Feed impression | random impressions | 43.6 | 8,347 |
| KuaiRec | 7,176 / 10,728 | 10.3M (13.4%) | Impression | fully observed 1,411 × 3,327 | 3,314 | 1,411 |

Coat (Schnabel et al., 2016) has 290 users and 300 items. The log holds 6,960 self-selected ratings (8.0% of the matrix), of which 1,905 are ratings of 4 or more. The test set holds 16 ratings per user on uniformly random items. Candidates are the items a user did not rate in the log, 14.7 per user on average. The dataset ships propensities, which its authors estimated from a uniform sample; Section 3 describes how they are used.

Yahoo! R3 (Marlin and Zemel, 2009) has 15,400 users and 1,000 songs. The log holds 311,704 self-selected ratings (2.0% dense). The test set holds 54,000 ratings of ten random songs for 5,400 users, in the version distributed with (Chen et al., 2021). A fixed random 5% of these users (270) is excluded from validation and testing. The remaining 5,130 test users have ten candidates each, and 2,315 of them have a positive candidate.

KuaiRand-Pure (Gao et al., 2022b) is the only benchmark whose log records platform exposure. An impression is an item shown in the standard feed, and the outcome is a click. We keep 10,000 users who also received randomly exposed videos, and all 7,583 videos. The log holds 511,863 impressions (0.68% dense), 47% of them clicks. The test set consists of the randomly exposed impressions of the same period, restricted to videos that the user was not shown in the standard log. Users have 43.6 candidates on average, and 8,347 of them have a positive candidate.

KuaiRec (Gao et al., 2022a) provides a dense log of 10.3 million views by 7,176 users of 10,728 items (13.4% dense). Its test set is a fully observed matrix for 1,411 users and 3,327 items. Each user is ranked over the test items that the user never viewed in the log, 3,314 on average.

Positive outcomes are ratings of 4 or more (Coat, Yahoo! R3), a click (KuaiRand) and a mean watch ratio of at least 2 (KuaiRec). On Coat and Yahoo! R3, "exposure" means that the user chose to rate the item, so the propensity models self-selection and not the allocation of a platform. The candidate sets of these two datasets hold about ten to fifteen items, which limits the range that nDCG can span on them.

### *4.2. Training-free operators and trained baselines*

The training-free operators are computed from the log and need no training. Pop ranks items by their popularity in the log. Impute ranks by the additive imputation of Section 4.3 alone. Obs, EASE, GF-CF, IPS, DR, DRUP, EASE-DR and GF-CF-DR are further training-free baselines, with the configuration counts of Table 4. DR and DRUP are defined here because the ensemble of Appendix B uses DRUP. The doubly robust edge estimate is

$$W_{ui} = \hat{Y}_{ui} + \frac{O_{ui}(Y_{ui} - \hat{Y}_{ui})}{\bar{p}_{ui}}, \tag{2}$$

which is unbiased for $Y_{ui}$ under correct propensities. Let $C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$, where $d_u$ and $d_i$ are the row and column sums of $W$ or of $\hat{Y}$ (a choice within the search space), and let $\alpha$ be a search space value. DRUP scores a pair by its one-hop term and its three-hop term, which is a walk-corrected propagation on the weighted graph $C \circ W$:

$$S_{ui} = \frac{(C \circ W)_{ui}}{\overline{|C \circ W|}_u} + \beta \frac{T_{ui}}{\overline{|T|}_u}, \tag{3}$$

where $T$ is the three-hop term with the walks that reuse an edge replaced by their unbiased counterparts, and $\overline{|\cdot|}_u$ is the mean absolute value over the items of user $u$. DR adjacency is the same operator without the walk correction. The weight $\beta$ and the other search space values are chosen on validation data. The training-free search spaces are the products of the following values, as run by the repository. On Coat, the propensity floor takes five values, the imputation penalty three, the imputation source two, $\alpha$ three, the degree source two and $\beta$ eight values, which gives $5 \times 3 \times 2 \times 3 \times 2 \times 8 =$ 1,440 for DR and for DRUP. On Yahoo! R3, the floor takes two values, the imputation penalty one, $\alpha$ two, the control-variate weight of the DR estimate four, the imputation source two, the degree source two and $\beta$ eight values, which gives $2 \times 1 \times 2 \times 4 \times 2 \times 2 \times 8 = 512$. On KuaiRec, the search space has three floors, two values for each of the other terms except $\beta$, and the eight values of $\beta$, which gives 384.

The trained baselines are the following. Their search spaces are given per dataset, and the learning rate and weight decay are shared by all trained models unless stated otherwise.

The per-dataset search spaces of the trained baselines, the AdvInfoNCE re-implementation and their search spaces are given in Appendix E.

### *4.3. The additive prior and the learners iALS-p and LightGCN-p*

*Exposure propensity..* On Yahoo! R3, KuaiRand-Pure and KuaiRec the exposure propensity is a logistic model fitted to the exposure matrix of the log,

$$\hat{p}_{ui} = \sigma\big(a + b\,\tilde{\ell}_u + c\,\tilde{\ell}_i\big),$$

where $\tilde{\ell}_u$ and $\tilde{\ell}_i$ are standardised logits of the user and item exposure rates. The coefficients are fitted by Newton's method on all $mn$ pairs. On Coat, the shipped propensities are used instead. Every propensity is then truncated below at a floor $f$, which gives $\bar{p}_{ui} = \max\{\hat{p}_{ui}, f\}$. The floor is an argument of each run and is not part of any configuration search space. Its values are 0.05 for every LightGCN-p and QP-LightGCN run on all four datasets. The exceptions are the following. On Coat, iALS-p and DRGP-iALS use 0.1, the prior alone uses 0.1, and the DRGP-LGN learners use 0.1. On Yahoo! R3, the prior alone uses 0.1. On KuaiRand, the prior alone uses 0.02. On KuaiRec, iALS-p, the prior alone and the DRGP-LGN learners use 0.2. The DRGP prior alone was run at floors 0.05, 0.1 and 0.2 on Coat and at 0.1 and 0.2 on KuaiRec. The stored result files do not record a rule for these values. The

floors of the training-free operators are part of their search spaces: Coat 0.01 to 0.2, Yahoo! R3 0.1 and 0.2, and KuaiRec 0.05 to 0.2.

*Additive imputation..* The imputed outcome is $\hat{Y}_{ui} = \min\{1, \max\{0, \mu + b_u + b_i\}\}$. Its parameters are fitted to the logged outcomes $O \odot Y$, with weights $O/\bar{p}$, by 20 alternating closed-form ridge sweeps with penalty $\lambda = 5$. The fit uses the log only and is not cross-fitted, so the same log supplies $\hat{Y}$ and the training data.

*Prior..* Let $\ell_{ui} = \operatorname{logit} \hat{Y}^c_{ui}$, where $\hat{Y}^c$ is $\hat{Y}$ clipped to $[10^{-3}, 1 - 10^{-3}]$. The prior is the standardised logit $z_{ui} = (\ell_{ui} - \bar{\ell})/s_\ell$, where the mean and the standard deviation are taken over all $mn$ pairs. Within a user, the ranking by $z$ is the ranking by $b_i$, up to ties created by the clipping of $\hat{Y}$ to $[0, 1]$. The prior alone is a training-free scorer (one configuration), and it is reported in Table 6.

*iALS-p..* The iALS objective of Hu et al. (Hu et al., 2008) receives a fixed offset,

$$\sum_{u,i} c_{ui} \big(y_{ui} - \beta z_{ui} - x_u^\top y_i\big)^2 + \operatorname{reg} \left(\|x_u\|^2 + \|y_i\|^2\right),$$

where $c_{ui} = 1 + \alpha$ on logged positives and $c_{ui} = 1$ elsewhere. The search space is $\alpha \in \{10, 30, 100\}$, regularisation in $\{10, 100, 300\}$, $\beta \in \{0.1, 0.2, 0.3, 0.5, 1\}$ and width in $\{32, 64\}$, which gives 90 configurations on Coat. On KuaiRec the main-table row uses an eight-configuration search space, $\alpha \in \{10, 30\}$, regularisation in $\{100, 1000\}$, $\beta \in \{0.5, 1\}$ and width 64. The DRGP runs use a larger search space of 45 configurations on KuaiRec ($\alpha \in \{10, 30, 100\}$, regularisation in $\{10, 100, 300\}$, the five values of $\beta$, width 64). Its additive row is reported as a reference outside the main comparison (Table A.12).

*LightGCN-p..* The offset enters the BPR margin of the LightGCN learner, so that $s_{ui} - s_{uj} = f_\theta(u, i) - f_\theta(u, j) + \beta(z_{ui} - z_{uj})$, with $\beta \in \{0.5, 1\}$. The search spaces are as follows. On Coat, learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}, 10^{-3}\}$, width in $\{32, 64\}$ and depth in $\{1, 2, 3\}$, which gives 108 configurations. On Yahoo! R3, learning rate $3 \cdot 10^{-3}$, weight decay in $\{10^{-5}, 10^{-4}\}$, width in $\{64, 128\}$ and depth in $\{1, 2, 3\}$, which gives 24. On KuaiRand, learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}\}$, width 64 and depth 2, which gives 12. On KuaiRec, learning rate $3 \cdot 10^{-3}$, weight decay in $\{10^{-5}, 10^{-4}\}$, width in $\{64, 128\}$ and depth in $\{1, 2, 3\}$, which gives 24.

The prior variants of the anatomy analysis (the IPS-free fit, the popularity offset, the shuffled item term and the user-only offset) are defined in Appendix E.3.

## 5. Results

This section answers the four research questions in turn. Test scores are nDCG on uniformly exposed test data, and Table 4 gives the benchmark with the number of configurations searched for each method. Paired differences are per-user averages of the difference between two methods, taken over the users who appear in both test sets. They are given in units of $10^{-3}$ nDCG, with the 95% confidence-interval half-width in brackets. A difference is called significant only when its Holm-adjusted p-value is below 0.05, and the Holm family is stated wherever a p-value is given. The analysis uses 225 users on Coat, 2,310 on Yahoo! R3, 8,321 on KuaiRand-Pure and 1,410 on KuaiRec. There are 10 splits on Coat and 5 on the other three datasets.

Table 5 reports the primary comparisons. P1 compares the strongest third-party graph debiaser with LightGCN, and P2 compares a simple debiased prior with its backbone; both are reported on all four datasets. P3 compares the best refinement with the simple prior of the same learner (Coat and KuaiRec), and P4 compares training-free doubly robust propagation with the best trained model on KuaiRec. The equivalence (TOST) margins are sensitivity thresholds chosen before the analysis and are not derived from the literature.

### *5.1. Effects of evaluation choices (RQ1)*

Selecting on the test split inflates the scores of the trained methods by a mean of +0.0207 on Coat (range +0.0097 to +0.0302), +0.0044 on Yahoo! R3, +0.0012 on KuaiRand-Pure and +0.0009 on KuaiRec. On Coat this inflation changes an order: under validation selection iALS scores 0.5812 and LightGCN 0.5683, but under the per-split test oracle, which chooses on test data and is a diagnostic only, they score 0.5909 and 0.5924, so the order reverses. The configuration-level oracle keeps the validation order. The sensitivity to the size of the searched space, the weighting unit, the metric and the seed comparisons is reported in Appendix C.

### *5.2. What graph debiasing adds over an equally tuned backbone (RQ2)*

On Coat the three graph debiasers with stored results differ from LightGCN by +3.76 [5.74] (r-AdjNorm), −1.59 [10.13] (NAVIP) and −0.65 [30.16]

Table 4: Main benchmark: mean test score over splits (sd over splits, ddof 0) and configuration count per method. Metric: nDCG@5 (Coat, Yahoo), nDCG@10 (KuaiRand), nDCG@20 (KuaiRec).

| Method | Coat | | Yahoo | | KuaiRand | | KuaiRec | |
|---|---|---|---|---|---|---|---|---|
| | mean (sd) | cfg | mean (sd) | cfg | mean (sd) | cfg | mean (sd) | cfg |
| Obs | 0.5571 (0.0100) | 24 | 0.6585 (0.0058) | 16 | – | – | 0.6036 (0.0028) | 16 |
| Pop | 0.5367 (0.0095) | 1 | 0.5331 (0.0059) | 1 | – | – | 0.5766 (0.0023) | 1 |
| Impute | 0.5532 (0.0097) | 6 | 0.5162 (0.0044) | 2 | – | – | 0.6194 (0.0018) | 4 |
| IPS | 0.5510 (0.0224) | 720 | – | – | – | – | – | – |
| DR | 0.5459 (0.0244) | 1440 | 0.6567 (0.0063) | 512 | – | – | 0.6378 (0.0020) | 384 |
| DRUP | 0.5467 (0.0235) | 1440 | 0.6567 (0.0063) | 512 | – | – | 0.6378 (0.0019) | 384 |
| EASE | 0.5401 (0.0138) | 3 | – | – | – | – | – | – |
| GF-CF | 0.5512 (0.0124) | 4 | – | – | – | – | – | – |
| EASE-DR | 0.5564 (0.0141) | 90 | – | – | – | – | – | – |
| GF-CF-DR | 0.5598 (0.0084) | 120 | – | – | – | – | – | – |
| MF | 0.4780 (0.0168) | 18 | 0.5087 (0.0075) | 9 | 0.3705 (0.0008) | 6 | 0.6221 (0.0036) | 6 |
| IPS-MF | 0.4713 (0.0160) | 18 | 0.5125 (0.0059) | 9 | – | – | 0.6252 (0.0018) | 6 |
| DR-MF | 0.5237 (0.0111) | 18 | 0.5093 (0.0042) | 9 | 0.3608 (0.0067) | 6 | 0.6219 (0.0025) | 6 |
| BPR-MF | 0.5174 (0.0206) | 18 | 0.6461 (0.0042) | 9 | 0.4475 (0.0027) | 6 | 0.5963 (0.0028) | 6 |
| PDA | 0.5181 (0.0138) | 36 | 0.6514 (0.0051) | 18 | – | – | – | – |
| LightGCN | 0.5683 (0.0132) | 54 | 0.6708 (0.0061) | 36 | 0.4569 (0.0010) | 6 | 0.6085 (0.0028) | 36 |
| LightGCN-pt | 0.5389 (0.0107) | 54 | 0.5947 (0.0028) | 6 | 0.4061 (0.0014) | 6 | 0.6271 (0.0023) | 36 |
| r-AdjNorm | 0.5722 (0.0134) | 108 | 0.6742 (0.0047) | 72 | 0.4590 (0.0024) | 12 | 0.6083 (0.0031) | 12 |
| NAVIP | 0.5664 (0.0163) | 54 | 0.6701 (0.0039) | 36 | 0.4565 (0.0017) | 6 | 0.6186 (0.0024) | 6 |
| DR-LightGCN | 0.5724 (0.0205) | 54 | 0.5528 (0.0091) | 6 | – | – | 0.6212 (0.0071) | 6 |
| SimGCL | 0.5528 (0.0141) | 108 | 0.6348 (0.0048) | 12 | 0.4196 (0.0022) | 12 | 0.5637 (0.0031) | 12 |
| MACR | 0.5418 (0.0134) | 36 | 0.6221 (0.0038) | 18 | 0.4311 (0.0019) | 12 | 0.5816 (0.0033) | 12 |
| AdvInfoNCE-LGN | 0.5222 (0.0204) | 216 | 0.6477 (0.0048) | 16 | 0.4480 (0.0017) | 8 | 0.5776 (0.0036) | 8 |
| DR-JL | 0.4811 (0.0164) | 18 | 0.5225 (0.0043) | 9 | 0.3759 (0.0021) | 6 | 0.6332 (0.0026) | 6 |
| MRDR | 0.4869 (0.0187) | 18 | 0.5248 (0.0059) | 9 | – | – | 0.6328 (0.0020) | 6 |
| iALS | 0.5812 (0.0101) | 24 | 0.6625 (0.0043) | 12 | 0.4528 (0.0017) | 12 | 0.6133 (0.0027) | 12 |
| iALS (search B) | 0.5869 (0.0118) | 18 | – | – | – | – | – | – |
| iALS-p (add prior) | 0.5984 (0.0121) | 90 | – | – | – | – | 0.6273 (0.0016) | 8 |
| LightGCN-p (add prior) | 0.6020 (0.0148) | 108 | 0.6690 (0.0038) | 24 | 0.4565 (0.0029) | 12 | 0.6288 (0.0020) | 24 |
| DR-iALS | 0.5871 (0.0113) | 48 | – | – | – | – | – | – |
| DR-BPR-LGN | 0.5912 (0.0122) | 108 | – | – | – | – | – | – |
| DR-BPR-LGN-1 (Q-only) | 0.5887 (0.0132) | 54 | – | – | – | – | – | – |
| DR-BPR-LGN-1 (Q-only) | – | – | 0.6837 (0.0057) | 36 | – | – | – | – |
| DR-BPR-LGN-p | 0.6039 (0.0139) | 108 | – | – | – | – | – | – |
| DR-BPR-LGN-p2 | 0.6046 (0.0177) | 108 | – | – | – | – | – | – |
| QP family (pooled) | 0.6015 (0.0162) | 324 | 0.6866 (0.0034) | 84 | 0.4613 (0.0024) | 30 | 0.6285 (0.0022) | 120 |
| control: shuffled Q | 0.5689 (0.0148) | 54 | 0.6689 (0.0054) | 12 | – | – | – | – |
| control: popularity Q | 0.5751 (0.0118) | 54 | 0.6550 (0.0046) | 12 | – | – | – | – |

Pooled rows (QP family) pool the configurations of their jobs; a cell marked – has no run for that method and dataset.

Table 5: Primary comparisons P1–P4 (Section 3): paired per-user differences, tests, MDE, TOST at the sensitivity margins (Section 3.5), and the matched-budget difference at $k = k_{\min}$.

| Dataset | Fam. | Comparison ($A - B$) | Diff. $\times 10^3$ [95% CI] | $p$ | $p_{\text{Holm}}$ | $p_{\text{NB,Holm}}$ | MDE $\times 10^3$ | TOST at margin | Matched-$k$ diff $\times 10^3$ | Seeds 1–3 diff $\times 10^3$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Coat | P1 | DR-LightGCN – LightGCN | -0.65 [30.16] | 0.966 | 1.000 | 1.000 | 43.07 | does not exclude $\lvert d\rvert > 0.010$ ($p = 0.271$) | +4.11 | – |
| Coat | P2 | iALS-p (add prior) – iALS | +16.82 [18.04] | 0.068 | 0.270 | 1.000 | 25.76 | does not exclude $\lvert d\rvert > 0.010$ ($p = 0.771$) | +16.12 | – |
| Coat | P2 | LightGCN-p (add prior) – LightGCN | +31.98 [20.65] | 0.003 | 0.013 | 1.000 | 29.48 | does not exclude $\lvert d\rvert > 0.010$ ($p = 0.981$) | +34.12 | – |
| Coat | P3 | QP family (pooled) – LightGCN-p (add prior) | +0.05 [8.29] | 0.991 | 1.000 | 1.000 | 11.83 | excludes $\lvert d\rvert > 0.010$ ($p = 0.009$) | -0.36 | – |
| Coat | P3 | DR-iALS – iALS-p (add prior) | -10.22 [15.96] | 0.208 | 0.625 | 1.000 | 22.79 | does not exclude $\lvert d\rvert > 0.010$ ($p = 0.511$) | -11.21 | – |
| Yahoo | P1 | r-AdjNorm – LightGCN | +4.10 [3.29] | 0.015 | 0.029 | 0.883 | 4.70 | yes ($p < 0.001$) | +2.87 | +5.26 ($p < 0.001$) |
| Yahoo | P2 | LightGCN-p (add prior) – LightGCN | -1.43 [3.93] | 0.475 | 0.475 | 0.883 | 5.62 | yes ($p < 0.001$) | -2.16 | – |
| KuaiRand | P1 | r-AdjNorm – LightGCN | +1.99 [1.02] | <0.001 | <0.001 | 1.000 | 1.46 | yes ($p = 0.026$) | +1.23 | +2.56 ($p < 0.001$) |
| KuaiRand | P2 | LightGCN-p (add prior) – LightGCN | -0.81 [1.33] | 0.234 | 0.234 | 1.000 | 1.90 | yes ($p < 0.001$) | -0.79 | – |
| KuaiRec | P1 | DR-LightGCN – LightGCN | +13.17 [3.31] | <0.001 | <0.001 | 0.787 | 4.74 | no ($p = 1.000$) | +12.97 | – |
| KuaiRec | P2 | LightGCN-p (add prior) – LightGCN | +21.22 [2.86] | <0.001 | <0.001 | 0.009 | 4.09 | no ($p = 1.000$) | +20.25 | – |
| KuaiRec | P2 | iALS-p (add prior) – iALS | +14.74 [3.43] | <0.001 | <0.001 | 0.043 | 4.90 | no ($p = 1.000$) | +16.89 | – |
| KuaiRec | P3 | QP family (pooled) – LightGCN-p (add prior) | -0.32 [0.50] | 0.211 | 0.211 | 0.871 | 0.72 | yes ($p < 0.001$) | -0.12 | – |
| KuaiRec | P4 | DRUP – DR-JL | +4.38 [2.21] | <0.001 | <0.001 | 0.426 | 3.15 | no ($p = 0.890$) | -14.90 | – |

Margins are sensitivity thresholds chosen before the analysis and not derived from the literature (Section 3.5). On Coat the TOST column states the differences the data exclude; equivalence is not claimed there. Seeds column: only comparisons whose baselines were retrained with seeds 1–3 (Section 3.4) are shown.

(DR-LightGCN), and none of these is significant. The TOST analysis at $\pm 0.010$ excludes differences larger than the margin for r-AdjNorm (TOST $p = 0.017$), but not for NAVIP ($p = 0.0515$) or DR-LightGCN ($p = 0.271$). The minimum detectable difference of P1 on Coat is 43.07, so the study cannot exclude a gain of that size for DR-LightGCN. The 95% half-widths of the three debiasers range from $\pm 6$ (r-AdjNorm) to $\pm 30$ (DR-LightGCN).

On Yahoo! R3, r-AdjNorm is +4.10 [3.29] above LightGCN ($p_{\text{Holm}} = 0.029$ in the two-comparison primary family; $p_{\text{Holm}} = 0.073$ over all comparisons with LightGCN in Table A.10; $p_{\text{NB,Holm}} = 0.883$ under the Nadeau-Bengio correction). On KuaiRand-Pure, r-AdjNorm is +1.99 [1.02] above LightGCN ($p_{\text{Holm}} < 0.001$), and this difference lies inside the sensitivity margin of $\pm 0.003$ (TOST $p = 0.026$), so the margin does not make it a null result. Its $p_{\text{NB,Holm}}$ is 1.000. Across Coat, Yahoo! R3 and KuaiRand-Pure, then, the best third-party graph debiaser adds at most $+4.10 \times 10^{-3}$ nDCG over the equally tuned backbone in the point estimate, and the Nadeau-Bengio correction removes its significance. The other third-party debasers with stored Yahoo! R3 results are below LightGCN: MACR by $-48.31$, SimGCL by $-36.49$ and AdvInfoNCE by $-21.08$ [7.78], each with $p_{\text{Holm}} < 0.05$. The bound applies to third-party debiasers only. Our own QP-LightGCN family adds +16.77 [5.82] over LightGCN on Yahoo! R3 ($p_{\text{Holm}} < 0.001$; +20.02 over seeds 1 to 3) and +3.93 [1.53] on KuaiRand-Pure ($p_{\text{Holm}} < 0.001$; +4.87 over seeds 1 to 3). Under the Nadeau-Bengio correction neither difference is significant ($p_{\text{NB,Holm}} = 0.434$ and 1.000). The refinements of the prior are reported as members of the evaluated family in Appendix B.

On KuaiRec, whose test matrix is fully observed, DR-LightGCN is +13.17 [3.31] above LightGCN ($p_{\text{Holm}} < 0.001$), and NAVIP is +10.41 [1.82] above it (Table A.12). Under the Nadeau-Bengio correction the DR-LightGCN difference is not significant ($p_{\text{NB,Holm}} = 0.787$), but the NAVIP difference ($p_{\text{NB,Holm}} = 0.034$) remains significant. The pointwise LightGCN-pt, which has no debiasing, is +19.59 [4.55] above the BPR backbone and also remains significant under the correction ($p_{\text{NB,Holm}} = 0.030$). The pointwise difference is a difference of means: no paired test compares it with the +13.17 of DR-LightGCN, and the two intervals overlap. The training-free operators are stronger on this dataset (Section 5.4).

The re-implementation of AdvInfoNCE is below LightGCN on all four datasets, by −46.47 [25.16] on Coat, −8.86 [2.50] on KuaiRand-Pure and −28.42 [6.46] on KuaiRec, each significant under the Holm adjustment of the backbone family (Tables A.9 to A.12). SimGCL and MACR are below LightGCN on all four datasets and significantly so on Yahoo! R3, KuaiRand-Pure and KuaiRec. On Coat, their differences (−17.49 [23.88] and −29.43 [24.60]) are not significant. These comparisons use our implementations, described in Appendix E.2.

Under the Nadeau-Bengio correction none of the four P1 comparisons is significant. The two P2 comparisons on KuaiRec remain significant, with $p_{\text{NB,Holm}}$ of 0.009 for LightGCN-p and 0.043 for iALS-p. The P4 comparison, DRUP against DR-JL, is significant under the per-user test (+4.38 [2.21], $p_{\text{Holm}} < 0.001$) but not under the correction ($p_{\text{NB,Holm}} = 0.426$).

### *5.3. Which ingredient carries the gain (RQ3)*

The simple prior is the additive imputation $\hat{\mu} + b_u + b_i$, fitted once on the exposed log with inverse-propensity weights and added to the score as an offset. Within a user its ranking is the ranking of the item term $b_i$. Used as an offset on LightGCN (LightGCN-p), it gives +31.98 [20.65] on Coat ($p_{\text{Holm}} = 0.013$ in the primary family of Table 5, $p_{\text{Holm}} = 0.061$ over all comparisons with LightGCN in Table A.9, and $p_{\text{NB,Holm}} = 1.000$ under the Nadeau-Bengio correction), and +21.22 [2.86] on KuaiRec ($p_{\text{Holm}} < 0.001$; $p_{\text{NB,Holm}} = 0.009$). On Yahoo! R3 it gives −1.43 [3.93] and on KuaiRand-Pure −0.81 [1.33], neither significant. The offset helps on Coat and KuaiRec and not on Yahoo! R3 or KuaiRand-Pure, and the tests here do not explain this pattern.

Table 6 separates the additive prior into its parts. Removing the inverse-propensity weights does not reduce the gain. On KuaiRec the unweighted

prior added to iALS gives +22.79 [4.04] over iALS, which exceeds the weighted prior by +8.05 [2.06] ($p_{\text{Holm}} < 0.001$). On Coat the unweighted prior is above the weighted one by +6.27 [6.97] for iALS and +4.01 [10.01] for LightGCN, neither significant. Popularity does not reproduce the effect on KuaiRec: its offset adds +1.06 [2.46] to iALS ($p_{\text{Holm}} = 1.000$), against +14.74 [3.43] for the weighted prior. Replacing the item term by a random permutation across items removes the KuaiRec gain and reverses it, with −112.07 [9.39] against iALS ($p_{\text{Holm}} < 0.001$). On KuaiRec the lowest offset weight of the search space was selected in all five splits for this control, so the reported harm is an upper bound. On Coat the same permutation gives −14.06 [18.53] for iALS and −12.95 [24.18] for LightGCN, neither significant, so the Coat result does not isolate the item term.

The weights on the sampled negatives are a second ingredient, and their support differs by dataset. On Yahoo! R3 the weights without the offset (DR-BPR-LGN-1) exceed LightGCN-p by +15.15 [5.84] ($p_{\text{Holm}} < 0.001$; $p_{\text{NB,Holm}} = 0.031$). A shuffled Q removes the gain of the weighted learner over the backbone, with −1.01 [3.20] against LightGCN ($p_{\text{Holm}} = 0.534$), and popularity weights are significantly below LightGCN (−15.91 [6.06], $p_{\text{Holm}} < 0.001$). On Coat, shuffled Q gives +0.97 [5.50] against LightGCN ($p_{\text{Holm}} = 0.73$), while popularity weights give +6.91 [6.04] ($p_{\text{Holm}} = 0.0505$, not significant). The QP family exceeds LightGCN-p on Yahoo! R3 by +18.20 [5.64] ($p_{\text{Holm}} < 0.001$; $p_{\text{NB,Holm}} = 0.002$) and on KuaiRand-Pure by +4.74 [1.54] ($p_{\text{Holm}} < 0.001$; $p_{\text{NB,Holm}} = 0.054$). No shuffled control was run on KuaiRand-Pure, and the validated combination selected $\beta = 0$ (no offset) in four of five splits, so the gain there is not from the offset in those splits. The weight claim is therefore supported by the shuffled control on Yahoo! R3 and is not established on Coat, KuaiRec or KuaiRand-Pure.

The refinements built on the prior do not improve on the simple prior of the same learner on Coat and KuaiRec, and they exceed it on Yahoo! R3 and KuaiRand-Pure (Table B.14 in Appendix B). DRGP, a leave-edge-out prior, was run on Coat and KuaiRec only, as single runs without seed repeats (10 of 10 and 5 of 5 jobs completed). The prior alone, without a learner, is reported in Table 6 and in Appendix B.

Two checks concern the inputs of the prior (Appendix C). Refitting the Coat prior rows with propensities estimated from the log changes the iALS-p and LightGCN-p gains by +2.63 [5.27] and +1.64 [10.30], and neither difference is significant. A wider iALS search space on Coat raises the stored iALS by +12.92 [8.01] ($p_{\text{Holm}} = 0.003$), a difference of the same order as the

Table 6: Prior anatomy. Test nDCG of iALS-p and LightGCN-p under four additive-imputation variants, and of the prior alone on all four datasets.

| Dataset | Learner / prior | nDCG (mean ± sd) | Δ vs no prior $\times 10^3$ [CI] | $p_{\text{Holm}}$ | Δ vs add $\times 10^3$ [CI] | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|
| Coat | iALS (no prior) | 0.5812 ± 0.0107 | – | – | – | – |
| Coat | iALS-p, add (IPS-fitted) | 0.5984 ± 0.0128 | +16.82 [18.04] | 0.811 | – | – |
| Coat | iALS-p, add, no IPS | 0.6044 ± 0.0115 | +23.08 [18.24] | 0.187 | +6.27 [6.97] | 0.856 |
| Coat | iALS-p, pop | 0.5832 ± 0.0238 | +1.67 [16.11] | 1.000 | -15.15 [15.39] | 0.698 |
| Coat | iALS-p, shuffled item bias | 0.5683 ± 0.0175 | -14.06 [18.53] | 1.000 | -30.87 [22.75] | 0.120 |
| Coat | LightGCN (no prior) | 0.5683 ± 0.0139 | – | – | – | – |
| Coat | LightGCN-p, add (IPS-fitted) | 0.6020 ± 0.0156 | +31.98 [20.65] | 0.046 | – | – |
| Coat | LightGCN-p, add, no IPS | 0.6052 ± 0.0145 | +36.00 [16.88] | <0.001 | +4.01 [10.01] | 1.000 |
| Coat | LightGCN-p, pop | 0.5835 ± 0.0233 | +15.63 [10.68] | 0.069 | -16.36 [19.52] | 1.000 |
| Coat | LightGCN-p, shuffled item bias | 0.5527 ± 0.0095 | -12.95 [24.18] | 1.000 | -44.93 [29.53] | 0.051 |
| KuaiRec | iALS (no prior) | 0.6133 ± 0.0030 | – | – | – | – |
| KuaiRec | iALS-p, add (IPS-fitted) | 0.6273 ± 0.0018 | +14.74 [3.43] | <0.001 | – | – |
| KuaiRec | iALS-p, add, no IPS | 0.6350 ± 0.0021 | +22.79 [4.04] | <0.001 | +8.05 [2.06] | <0.001 |
| KuaiRec | iALS-p, pop | 0.6144 ± 0.0028 | +1.06 [2.46] | 1.000 | -13.68 [3.31] | <0.001 |
| KuaiRec | iALS-p, shuffled item bias | 0.4993 ± 0.0032 | -112.07 [9.39] | <0.001 | -126.82 [9.29] | <0.001 |
| KuaiRec | LightGCN (no prior) | 0.6085 ± 0.0031 | – | – | – | – |
| KuaiRec | LightGCN-p, add (IPS-fitted) | 0.6288 ± 0.0023 | +21.22 [2.86] | <0.001 | – | – |
| Coat | prior alone, add (IPS-fitted) | 0.5585 ± 0.0105 | – | – | – | – |
| Coat | prior alone, add, no IPS | 0.5574 ± 0.0107 | – | – | -2.09 [8.46] | 1.000 |
| Coat | prior alone, pop | 0.5367 ± 0.0100 | – | – | -19.40 [27.79] | 1.000 |
| Coat | prior alone, shuffled item bias | 0.3572 ± 0.0125 | – | – | -203.62 [51.90] | <0.001 |
| Coat | prior alone, user only (b_i = 0) | 0.3497 ± 0.0152 | – | – | -207.15 [49.91] | <0.001 |
| Yahoo | prior alone, add (IPS-fitted) | 0.5243 ± 0.0050 | – | – | – | – |
| Yahoo | prior alone, add, no IPS | 0.5237 ± 0.0055 | – | – | -1.09 [3.98] | 1.000 |
| Yahoo | prior alone, pop | 0.5331 ± 0.0065 | – | – | +8.90 [14.93] | 1.000 |
| Yahoo | prior alone, shuffled item bias | 0.3446 ± 0.0056 | – | – | -177.98 [17.37] | <0.001 |
| Yahoo | prior alone, user only (b_i = 0) | 0.3767 ± 0.0051 | – | – | -144.74 [17.38] | <0.001 |
| KuaiRand | prior alone, add (IPS-fitted) | 0.3637 ± 0.0013 | – | – | – | – |
| KuaiRand | prior alone, add, no IPS | 0.3837 ± 0.0023 | – | – | +19.90 [2.45] | <0.001 |
| KuaiRand | prior alone, pop | 0.4204 ± 0.0021 | – | – | +56.41 [5.19] | <0.001 |
| KuaiRand | prior alone, shuffled item bias | 0.3257 ± 0.0018 | – | – | -39.12 [5.64] | <0.001 |
| KuaiRand | prior alone, user only (b_i = 0) | 0.3259 ± 0.0011 | – | – | -38.84 [5.53] | <0.001 |
| KuaiRec | prior alone, add (IPS-fitted) | 0.6216 ± 0.0016 | – | – | – | – |
| KuaiRec | prior alone, add, no IPS | 0.6367 ± 0.0022 | – | – | +15.16 [1.68] | <0.001 |
| KuaiRec | prior alone, pop | 0.5766 ± 0.0026 | – | – | -45.70 [4.25] | <0.001 |
| KuaiRec | prior alone, shuffled item bias | 0.0398 ± 0.0007 | – | – | -582.01 [10.19] | <0.001 |
| KuaiRec | prior alone, user only (b_i = 0) | 0.0686 ± 0.0010 | – | – | -554.01 [10.16] | <0.001 |

Δ = per-user paired difference (mean over splits), $\times 10^3$, [half-width of the 95% CI]. No-prior references: stored iALS and LightGCN. The add row is the IPS-fitted additive prior. $p_{\text{Holm}}$ over all paired tests of this table. user only (b_i = 0) gives the same score to every item for a given user, so its ranking is decided by tie-breaking. Coat LightGCN-p and iALS-p use 10 splits, KuaiRec 5. Rows absent from the table were not run (only the prior-alone rows exist for those priors).

gain of the offset model over the same stored iALS (+16.82 [18.04]). The choice of search space therefore moves the backbone by an amount of the same order as the prior, and the two cannot be separated by the present tests.

### *5.4. Exact unbiasedness of the propagation (RQ4)*

The walk correction removes the repeated-edge terms of multi-hop doubly robust propagation. On the three real logs on which DR and DRUP were run, it does not help at matched configurations. On Coat, DRUP minus DR is negative at every value of $\beta$ in the search space, from $-0.33$ at $\beta = 0.03$ to $-8.17$ at $\beta = 1000$, and it is negative in all ten splits for $\beta \geq 0.3$. On Yahoo! R3 the differences are small and negative, from $-0.02$ to $-0.32$. In the selected runs, the difference is +1.40 [3.86] on Coat ($p_{\text{Holm}} = 0.475$), where the data exclude differences larger than 0.010 in absolute value and equivalence is not claimed; it is $-0.12$ [0.15] on KuaiRec ($p_{\text{Holm}} = 0.113$, within $\pm 0.003$); and it is identically zero on Yahoo! R3, where the two selected runs coincide, so that row is uninformative. Table 7 gives the values. DR and DRUP were not run on KuaiRand-Pure.

A semi-synthetic KuaiRec study with known true labels gives a different picture, which is reported in Appendix D. Its nDCG@20 on unexposed candidates is higher for DRUP than for DR in seven of nine cells (Table D.18), and its relative bias for the three-hop term is lower for DRUP in eight of nine cells (Table D.19). The exceptions are the two imputed-weight cells with an independent or a 20% split nuisance fit, where DR has the higher nDCG, and the cross-fitted imputed-weight cell, where DRUP has the larger bias. This setting differs from the real logs, and we do not claim that the gain carries over to them. Exact unbiasedness therefore does not improve ranking on the three real logs on which DR and DRUP were run.

## 6. Discussion

### *6.1. What the evidence supports*

Table 8 states the answer to each research question and the strength of the evidence behind it. The answers are those of the introduction. The strength column records the design features that limit each claim. The search space chosen for a backbone can move its score by an amount comparable to the gain of a prior, so budgets have to be reported and matched before a gap is attributed to a method.

Table 7: Walk correction (DRUP minus DR) at matched configurations and in the selected runs.

| Dataset | $\beta$ | Mean diff. $\times 10^3$ [95% CI] | Share of configs $< 0$ | Splits with mean $< 0$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|
| *Matched configurations (split means, no test)* | | | | | |
| Coat | 0.03 | -0.33 | 0.49 | 7/10 | – |
| Coat | 0.1 | -1.28 | 0.52 | 9/10 | – |
| Coat | 0.3 | -2.92 | 0.54 | 10/10 | – |
| Coat | 1 | -5.09 | 0.74 | 10/10 | – |
| Coat | 3 | -7.35 | 0.74 | 10/10 | – |
| Coat | 10 | -7.44 | 0.80 | 10/10 | – |
| Coat | 1000 | -8.17 | 0.90 | 10/10 | – |
| Yahoo | 0.03 | -0.02 | 0.41 | 5/5 | – |
| Yahoo | 0.1 | -0.03 | 0.50 | 4/5 | – |
| Yahoo | 0.3 | -0.12 | 0.56 | 5/5 | – |
| Yahoo | 1 | -0.04 | 0.47 | 3/5 | – |
| Yahoo | 3 | -0.07 | 0.28 | 4/5 | – |
| Yahoo | 10 | -0.19 | 0.41 | 5/5 | – |
| Yahoo | 1000 | -0.32 | 0.38 | 5/5 | – |
| *Selected runs (per-user paired test)* | | | | | |
| Coat | – | +1.40 [3.86] | – | – | 0.475 |
| Yahoo | – | +0.00 [0.00] | – | – | 1.000 |
| KuaiRec | – | -0.12 [0.15] | – | – | 0.113 |

Panel A: difference of the matched-configuration split means, $\times 10^3$; the last column is the number of splits with a negative mean. Panel B: per-user differences with 95% half-widths in brackets, and Holm-adjusted $p$ within the walk family. Coat: differences larger than $\pm 0.010$ are excluded (TOST $p < 0.001$); equivalence is not claimed. On Yahoo! R3 the selected DRUP and DR runs give identical per-user values, so the walk correction changes nothing there and the row is uninformative. KuaiRec: TOST within the sensitivity margin ($\pm 0.003$), $p < 0.001$.

Table 8: Answers to the research questions and the strength of the evidence for each.

| Question | Answer | Strength of the evidence |
|---|---|---|
| RQ1 | Evaluation choices change the conclusions. Test-set selection inflates the scores and reverses one order on Coat, and many pairwise signs depend on the search budget or the metric. | Oracle and subset diagnostics are descriptive. The Coat half-widths are large, so the Coat ordering is the least certain result. |
| RQ2 | Third-party graph debiasers add little over an equally tuned backbone on Coat, Yahoo! R3 and KuaiRand-Pure. On KuaiRec the gains are larger, and a pointwise loss without debiasing gives a larger difference. | Holm-adjusted paired tests on four datasets. The Nadeau-Bengio correction removes significance from every primary comparison of a third-party debiaser. Coat is underpowered. |
| RQ3 | The additive prior gives the largest point-estimate gains over LightGCN on Coat and KuaiRec. Its inverse-propensity weights are not what matters in our runs, and the mechanism is not identified. | Controls without weights, with popularity, and with a shuffled item term, on two datasets. The Coat controls are underpowered, and the item-term reading is a hypothesis. |
| RQ4 | The walk correction does not improve ranking on Coat, Yahoo! R3 or KuaiRec. | Matched-configuration and selected-run comparisons on the three real logs. The semi-synthetic study is a different setting and is not evidence about the real logs. |

*6.2. Implications for evaluating debiasing methods*

The results support five practices.

- **Keep uniform data out of training, and state where every nuisance comes from.** When the propensities are not fitted on the training log, report their source and refit the rows that depend on them with the estimator used elsewhere.
- **Select per split on validation data, and report the gap to the test oracle.** Publish the per-split validation and test values, because the inflation, the subset stability and the bootstrap ranks can be recomputed only from them.
- **Report configuration counts and compare at matched budget.** Give the search space of every method, and widen the search for the strongest baseline before attributing a gap to the proposed method.
- **Include simple baselines as mandatory rows.** These are the IPS imputation alone, a pointwise backbone, a popularity ranker, the prior alone, and the backbone with the debiased prior.
- **Retrain the selected configurations with several seeds, and report corrected tests next to standard ones.** State the minimum detectable difference with every claim of no difference, and test equivalence only with a margin fixed in advance.

*6.3. Limitations and threats to validity*

1. **Power and sensitivity margins.** Coat has 225 evaluated users, and its minimum detectable differences bound what the data can show. Coat results are therefore statements of no detectable difference, together with the exclusion of differences above the stated margin, and never statements of equivalence. The TOST margins ($\pm 0.010$ on Coat and Yahoo! R3, $\pm 0.003$ on KuaiRand-Pure and KuaiRec) are sensitivity thresholds chosen before the analysis, not derived from the literature.
2. **Implementations.** The graph baselines NAVIP, r-AdjNorm, SimGCL and MACR, the refinements, and the AdvInfoNCE re-implementation are our own. The re-implementation simplifies the adversary budget, has no KL-ball projection, and uses different search spaces across

datasets. The official AdvInfoNCE code and the PPD post hoc method were not run, so the negative results for these methods concern our implementations under this protocol. The Yahoo! R3 DR-LightGCN row (0.5528, 6 configurations) lies far below the other graph models and may reflect under-tuning or an implementation limit.

3. **Refinement family.** The refinements of the prior were developed as candidate methods, and they are reported in full with their configuration counts. The family is not exhaustive, and the results do not show that no other refinement helps.
4. **Seeds, budgets and DRGP.** Seeds 1 to 3 were not run for the Coat prior rows. The seed checks cover the leading methods only, and their baselines were chosen on test data, so they are descriptive. DRGP has single runs on Coat and KuaiRec. Budgets are unequal. Matched-budget differences are split means without tests, and the subset analysis draws from the stored configurations rather than running new searches. On Coat, the choice of search space moved iALS by +12.92 [8.01] ($p_{\text{Holm}} = 0.003$), an amount of the same order as the gain of the prior, so budget alone can move a baseline by that much.
5. **Search-space edges and ablation coverage.** On KuaiRec the lowest offset weight of the search space was selected in all five splits for the shuffled item term, so the reported harm is an upper bound under that search space. The unweighted prior selects the largest regularisation value of its search space (1000) on KuaiRec, and the wider iALS search space is not exhausted. The user-only prior was run only as a prior alone. LightGCN with estimated propensities was not run, and on KuaiRec only the additive prior was run in the offset-only LightGCN setting.
6. **Statistical dependence and family choice.** Per-user tests over overlapping splits treat the log and the fitted nuisances as fixed, and they can be anti-conservative. The Nadeau-Bengio correction is a heuristic (Section 3.5), and it removes significance from every primary comparison of a third-party debiaser. Several p-values change with the Holm family. The Yahoo! R3 r-AdjNorm comparison has $p_{\text{Holm}} = 0.029$ in the primary family and 0.073 over all comparisons with LightGCN, and the Coat offset comparison has 0.013 in the primary family, 0.046 in the family of Table 6, and 0.061 over all comparisons with LightGCN. Each value is reported with its family. The reconciliation of the stored results recomputes 151 of 154 method means within 0.002, and

the three that do not agree are used as stored (Appendix E.4).

7. **Exposure semantics and test design.** On Coat and Yahoo! R3, exposure is user self-selection, and the weights assume that exposure is independent of the outcome given the covariates of the propensity model. One cut-off is used per dataset. KuaiRec has a fully observed test matrix with about 3,300 candidates per user, unlike the other three datasets, and we have no test of which feature causes the difference.
8. **Synthetic evidence and mechanism.** The bias results for the walk correction come from a synthetic MNAR model and a semi-synthetic KuaiRec study. Neither measures ranking on real logs, and the semi-synthetic nDCG results do not carry over to the real logs at matched configurations. The item-term reading of the prior is a hypothesis. The results show that an item-level term fitted once on the exposed log carries the effect on KuaiRec, but not why. The Impute row and the prior-alone row on KuaiRec differ in propensity floor and penalty (Section 5.3), and no paired test compares the prior alone with DRUP.

## 7. Conclusion

We re-evaluated exposure-bias corrections for graph recommenders on four benchmarks with uniformly exposed test data, under one protocol in which uniform data serve for validation and testing, and, through Coat's shipped propensities, for one nuisance that we check for the prior rows. The study asked how much evaluation choices move conclusions, how much graph-specific debiasing adds over an equally tuned backbone, which ingredient carries the gains that remain, and whether exact unbiasedness of the propagation matters for ranking.

First, the protocol is sensitive to evaluation choices. Selecting on test data inflates Coat scores by up to 0.030 nDCG@5, and under the per-split test oracle it reverses the order of iALS and LightGCN. The search space of a backbone can move a baseline by an amount comparable to the gain of a prior. Second, third-party graph-specific debiasers add at most $+4.10 \times 10^{-3}$ nDCG over LightGCN in the point estimates on Coat, Yahoo! R3 and KuaiRand-Pure, and none of these gains survives the correction for overlapping splits on those datasets. On KuaiRec the gains are larger, NAVIP's gain survives the correction, and a pointwise loss without any debiasing gives a larger mean difference. The QP-LightGCN family is not significant under the correction on Yahoo! R3 or KuaiRand-Pure. Third, an additive prior, fitted once

on the exposed log and added to the score, gives the largest point-estimate gains over LightGCN on Coat and KuaiRec. The Coat gain is not robust to the correction for overlapping splits, and the KuaiRec gain is. Removing the inverse-propensity weights does not reduce the gain, a shuffled item term removes the KuaiRec gain, and popularity does not reproduce it. The mechanism is not identified. Fourth, the walk correction, which makes multi-hop doubly robust propagation exactly unbiased, does not improve ranking on Coat, Yahoo! R3 or KuaiRec. In a semi-synthetic KuaiRec study with known labels, DRUP has the higher nDCG@20 in seven of nine cells. That setting differs from the real logs, and we do not take it to show that the correction helps there. DR and DRUP were not run on KuaiRand-Pure.

The study does not propose a method. It does not show that debiasing is ineffective in general, because its scope is four public benchmarks, one cut-off per dataset and offline evaluation of implicit-feedback learners trained on the biased log. It does not establish equivalence on Coat, where the study is underpowered. For evaluation, the practical requirements are to select per split on validation data, to match budgets, to include the simple baselines, and to report minimum detectable effects and corrected tests with every claim that two methods are equal or that one is better.

## Appendix A. Every method against the backbone

Tables A.9 to A.12 give, for each dataset, every method against LightGCN, which is the backbone family of Section 3.5.

## Appendix B. Refinement family

This appendix describes the four refinements of the additive prior that are reported as one evaluated family in Section 4, with the Q-weighted negatives, the QP-LightGCN family, the ensemble with DRUP and the leave-edge-out DR prior (DRGP). Their configuration counts are in Table B.13, and the paired tests of the refinements against the simple prior are in Table B.14.

### *Appendix B.1. Q-weighted negatives*

The probability that an unlogged pair is a positive is

$$Q_{ui} = \frac{\hat{Y}_{ui}(1 - \bar{p}_{ui})}{1 - \bar{p}_{ui}\hat{Y}_{ui}}, \tag{B.1}$$

Table A.9: Appendix: every method against LightGCN on Coat (backbone family, nDCG@5). Holm is taken over this family.

| Method | Diff. $\times 10^3$ [95% CI] | $p$ | $p_{\text{Holm}}$ | $p_{\text{NB,Holm}}$ | MDE $\times 10^3$ |
|---|---|---|---|---|---|
| MF | -93.59 [33.10] | <0.001 | <0.001 | 0.413 | 47.26 |
| IPS-MF | -98.45 [33.78] | <0.001 | <0.001 | 0.196 | 48.23 |
| DR-MF | -47.85 [32.93] | 0.005 | 0.106 | 0.196 | 47.02 |
| BPR-MF | -53.28 [21.08] | <0.001 | <0.001 | 1.000 | 30.10 |
| PDA | -50.81 [20.12] | <0.001 | <0.001 | 1.000 | 28.73 |
| LightGCN-pt | -32.48 [32.29] | 0.049 | 0.779 | 1.000 | 46.11 |
| r-AdjNorm | +3.76 [5.74] | 0.198 | 1.000 | 1.000 | 8.20 |
| NAVIP | -1.59 [10.13] | 0.758 | 1.000 | 1.000 | 14.47 |
| DR-LightGCN | -0.65 [30.16] | 0.966 | 1.000 | 1.000 | 43.07 |
| DR-JL | -91.00 [35.14] | <0.001 | <0.001 | 0.203 | 50.18 |
| MRDR | -85.16 [36.17] | <0.001 | <0.001 | 0.734 | 51.65 |
| MACR | -29.43 [24.60] | 0.019 | 0.405 | 1.000 | 35.13 |
| SimGCL | -17.49 [23.88] | 0.150 | 1.000 | 1.000 | 34.09 |
| iALS | +12.01 [16.12] | 0.143 | 1.000 | 1.000 | 23.01 |
| AdvInfoNCE-LGN | -46.47 [25.16] | <0.001 | 0.009 | 1.000 | 35.93 |
| iALS (search B) | +19.40 [16.48] | 0.021 | 0.425 | 1.000 | 23.53 |
| iALS-p (add prior) | +28.83 [20.48] | 0.006 | 0.132 | 1.000 | 29.25 |
| LightGCN-p (add prior) | +31.98 [20.65] | 0.003 | 0.061 | 1.000 | 29.48 |
| DR-BPR-LGN | +23.96 [8.67] | <0.001 | <0.001 | 1.000 | 12.38 |
| DR-BPR-LGN-1 (Q-only) | +21.20 [9.26] | <0.001 | <0.001 | 1.000 | 13.22 |
| DR-BPR-LGN-p | +34.07 [20.11] | <0.001 | 0.025 | 1.000 | 28.72 |
| DR-BPR-LGN-p2 | +35.26 [20.51] | <0.001 | 0.022 | 1.000 | 29.28 |
| DR-iALS | +18.61 [16.31] | 0.026 | 0.480 | 1.000 | 23.28 |
| QP family (pooled) | +32.03 [17.27] | <0.001 | 0.009 | 1.000 | 24.65 |
| control: shuffled Q | +0.97 [5.50] | 0.730 | 1.000 | 1.000 | 7.86 |
| control: popularity Q | +6.91 [6.04] | 0.025 | 0.480 | 1.000 | 8.63 |
| Obs | -12.94 [30.72] | 0.408 | 1.000 | 1.000 | 43.87 |
| EASE | -28.72 [32.13] | 0.080 | 1.000 | 1.000 | 45.88 |
| GF-CF | -18.32 [30.12] | 0.232 | 1.000 | 1.000 | 43.01 |
| IPS | -18.36 [31.48] | 0.252 | 1.000 | 1.000 | 44.96 |
| DR | -26.62 [32.00] | 0.103 | 1.000 | 1.000 | 45.69 |
| DRUP | -25.21 [32.81] | 0.131 | 1.000 | 1.000 | 46.85 |
| EASE-DR | -15.27 [33.69] | 0.373 | 1.000 | 1.000 | 48.10 |
| GF-CF-DR | -12.33 [32.46] | 0.455 | 1.000 | 1.000 | 46.35 |
| Pop | -34.67 [31.12] | 0.029 | 0.495 | 1.000 | 44.43 |
| Impute | -16.97 [34.67] | 0.336 | 1.000 | 1.000 | 49.50 |

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.5.

Table A.10: Appendix: every method against LightGCN on Yahoo (backbone family, nDCG@5). Holm is taken over this family.

| Method | Diff. $\times 10^3$ [95% CI] | $p$ | $p_{\text{Holm}}$ | $p_{\text{NB,Holm}}$ | MDE $\times 10^3$ |
|---|---|---|---|---|---|
| MF | -162.69 [15.25] | <0.001 | <0.001 | 0.021 | 21.80 |
| IPS-MF | -159.11 [15.20] | <0.001 | <0.001 | 0.012 | 21.72 |
| DR-MF | -162.10 [16.02] | <0.001 | <0.001 | 0.012 | 22.89 |
| BPR-MF | -26.53 [8.17] | <0.001 | <0.001 | 0.218 | 11.68 |
| PDA | -20.36 [7.73] | <0.001 | <0.001 | 0.021 | 11.05 |
| LightGCN-pt | -76.19 [13.05] | <0.001 | <0.001 | 0.061 | 18.65 |
| r-AdjNorm | +4.10 [3.29] | 0.015 | 0.073 | 1.000 | 4.70 |
| NAVIP | -0.74 [3.11] | 0.642 | 1.000 | 1.000 | 4.45 |
| DR-LightGCN | -117.88 [14.56] | <0.001 | <0.001 | 0.019 | 20.81 |
| DR-JL | -149.61 [14.97] | <0.001 | <0.001 | 0.016 | 21.39 |
| MRDR | -146.84 [14.97] | <0.001 | <0.001 | 0.006 | 21.40 |
| MACR | -48.31 [9.71] | <0.001 | <0.001 | 0.021 | 13.87 |
| SimGCL | -36.49 [9.61] | <0.001 | <0.001 | 0.042 | 13.74 |
| iALS | -8.11 [7.99] | 0.047 | 0.187 | 1.000 | 11.42 |
| AdvInfoNCE-LGN | -21.08 [7.78] | <0.001 | <0.001 | 0.135 | 11.12 |
| LightGCN-p (add prior) | -1.43 [3.93] | 0.475 | 1.000 | 1.000 | 5.62 |
| DR-BPR-LGN-1 (Q-only) | +13.72 [5.27] | <0.001 | <0.001 | 0.291 | 7.54 |
| QP family (pooled) | +16.77 [5.82] | <0.001 | <0.001 | 0.434 | 8.31 |
| control: shuffled Q | -1.01 [3.20] | 0.534 | 1.000 | 1.000 | 4.57 |
| control: popularity Q | -15.91 [6.06] | <0.001 | <0.001 | 1.000 | 8.66 |
| Obs | -12.40 [8.49] | 0.004 | 0.025 | 1.000 | 12.13 |
| DR | -14.58 [8.51] | <0.001 | 0.006 | 0.511 | 12.16 |
| DRUP | -14.58 [8.51] | <0.001 | 0.006 | 0.511 | 12.16 |
| Pop | -138.65 [13.06] | <0.001 | <0.001 | <0.001 | 18.67 |
| Impute | -155.70 [16.32] | <0.001 | <0.001 | 0.016 | 23.32 |

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.5.

Table A.11: Appendix: every method against LightGCN on KuaiRand (backbone family, nDCG@10). Holm is taken over this family.

| Method | Diff. $\times 10^3$ [95% CI] | $p$ | $p_{\text{Holm}}$ | $p_{\text{NB,Holm}}$ | MDE $\times 10^3$ |
|---|---|---|---|---|---|
| MF | -87.35 [5.38] | <0.001 | <0.001 | <0.001 | 7.69 |
| DR-MF | -96.98 [5.32] | <0.001 | <0.001 | 0.010 | 7.61 |
| BPR-MF | -10.06 [2.62] | <0.001 | <0.001 | 0.574 | 3.74 |
| LightGCN-pt | -52.46 [4.85] | <0.001 | <0.001 | 0.001 | 6.93 |
| r-AdjNorm | +1.99 [1.02] | <0.001 | <0.001 | 1.000 | 1.46 |
| NAVIP | -0.43 [1.14] | 0.461 | 0.468 | 1.000 | 1.63 |
| DR-JL | -82.06 [4.90] | <0.001 | <0.001 | <0.001 | 7.01 |
| MACR | -26.46 [3.58] | <0.001 | <0.001 | 0.012 | 5.12 |
| SimGCL | -38.27 [4.12] | <0.001 | <0.001 | 0.005 | 5.89 |
| iALS | -5.05 [3.16] | 0.002 | 0.005 | 1.000 | 4.51 |
| AdvInfoNCE-LGN | -8.86 [2.50] | <0.001 | <0.001 | 0.120 | 3.57 |
| LightGCN-p (add prior) | -0.81 [1.33] | 0.234 | 0.468 | 1.000 | 1.90 |
| QP family (pooled) | +3.93 [1.53] | <0.001 | <0.001 | 1.000 | 2.19 |

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.5.

clipped to $[0, 1]$. This is the posterior under the assumption that exposure and outcome are independent given the covariates on which the propensity depends.

The Q-weighted BPR loss uses only logged positives. For each positive $(u, i)$ one negative $j$ is drawn uniformly from the items, and it is redrawn once if $(u, j)$ is itself logged. The negative is weighted by $w_{uj} = \max\{1 - f_n Q_{uj}, 10^{-3}\}$, and the loss over a batch $B$ is

$$\mathcal{L} = -\frac{\sum_{(u,i,j)\in B} w_{uj} \log \sigma\big(s_{ui} - s_{uj}\big)}{\sum_{(u,i,j)\in B} w_{uj}} + \lambda \frac{1}{|B|} \sum_{(u,i,j)\in B} \big(\|e_u\|^2 + \|e_i\|^2 + \|e_j\|^2\big),$$

where $e_u$, $e_i$ and $e_j$ are the embeddings and $\lambda$ is the weight decay. The normalisation by the total weight keeps the loss scale independent of how many negatives are down-weighted. The factor $f_n$ is 1 in every job except the Coat job without offset, where $f_n \in \{0.5, 1\}$. The encoder is LightGCN.

The Q-weighted iALS (DR-iALS) uses the same weights as confidences. Logged positives get $c_{ui} = 1 + \alpha$, and unlogged pairs get $c_{ui} = \max\{1 - f_n Q_{ui}, 10^{-3}\}$. Its search space is $\alpha \in \{3, 10, 30\}$, regularisation in $\{1, 10, 100, 1000\}$, $f_n \in \{0.5, 1\}$ and width in $\{32, 64\}$, which gives 48 configurations on Coat.

The configuration counts of the Q-weighted LightGCN (DR-BPR-LGN) are 108 on Coat (with $f_n \in \{0.5, 1\}$), and DR-BPR-LGN-1 (with $f_n = 1$) has

Table A.12: Appendix: every method against LightGCN on KuaiRec (backbone family, nDCG@20). Holm is taken over this family.

| Method | Diff. $\times 10^3$ [95% CI] | $p$ | $p_{\text{Holm}}$ | $p_{\text{NB,Holm}}$ | MDE $\times 10^3$ |
|---|---|---|---|---|---|
| LightGCN-pt | +19.59 [4.55] | <0.001 | <0.001 | 0.030 | 6.51 |
| r-AdjNorm | +0.22 [0.80] | 0.585 | 0.585 | 0.877 | 1.14 |
| NAVIP | +10.41 [1.82] | <0.001 | <0.001 | 0.034 | 2.60 |
| DR-LightGCN | +13.17 [3.31] | <0.001 | <0.001 | 0.787 | 4.74 |
| BPR-MF | -11.73 [3.27] | <0.001 | <0.001 | 0.057 | 4.68 |
| SimGCL | -44.34 [4.11] | <0.001 | <0.001 | <0.001 | 5.88 |
| DR-JL | +25.46 [3.66] | <0.001 | <0.001 | 0.057 | 5.22 |
| MRDR | +25.24 [3.53] | <0.001 | <0.001 | 0.031 | 5.05 |
| MACR | -25.84 [4.57] | <0.001 | <0.001 | 0.018 | 6.53 |
| iALS | +4.98 [2.38] | <0.001 | <0.001 | 0.107 | 3.41 |
| MF | +14.63 [3.96] | <0.001 | <0.001 | 0.353 | 5.65 |
| IPS-MF | +17.69 [3.53] | <0.001 | <0.001 | 0.089 | 5.04 |
| DR-MF | +14.47 [4.20] | <0.001 | <0.001 | 0.124 | 6.00 |
| AdvInfoNCE-LGN | -28.42 [6.46] | <0.001 | <0.001 | 0.034 | 9.23 |
| LightGCN-p (add prior) | +21.22 [2.86] | <0.001 | <0.001 | 0.030 | 4.09 |
| iALS-p (add prior) | +19.72 [3.32] | <0.001 | <0.001 | 0.058 | 4.74 |
| iALS-p (add, DRGP grid) | +19.96 [2.71] | <0.001 | <0.001 | 0.016 | 3.87 |
| QP family (pooled) | +20.90 [2.88] | <0.001 | <0.001 | 0.089 | 4.11 |
| Obs | -4.80 [3.11] | 0.002 | 0.005 | 0.094 | 4.44 |
| DRUP | +29.84 [3.51] | <0.001 | <0.001 | 0.011 | 5.01 |
| DR | +29.96 [3.51] | <0.001 | <0.001 | 0.011 | 5.01 |
| Pop | -31.41 [3.25] | <0.001 | <0.001 | <0.001 | 4.64 |
| Impute | +11.89 [3.93] | <0.001 | <0.001 | 0.301 | 5.62 |

Per-user paired differences over the test users; the Nadeau-Bengio column is the heuristic correction of Section 3.5.

Table B.13: Configurations searched by the refinement family, its controls and the DRGP runs (stored counts).

| Member | Coat | Yahoo | KuaiRand | KuaiRec |
|---|---|---|---|---|
| DR-BPR-LGN (Q-weighted LightGCN) | 108 | – | – | – |
| DR-BPR-LGN-1 (Q-only) | 54 | 36 | 6 | 24 |
| DR-BPR-LGN-p (offset, $\beta \in \{0.5, 1\}$) | 108 | 24 | 12 | 24 |
| DR-BPR-LGN-p2 (offset, $\beta \in \{1, 2\}$) | 108 | 24 | 12 | 72 |
| DR-iALS | 48 | – | – | – |
| QP family (pooled over its jobs) | 324 | 84 | 30 | 120 |
| Control: shuffled Q | 54 | 12 | – | – |
| Control: popularity Q | 54 | 12 | – | – |
| DRGP prior alone | 6 | – | – | 2 |
| DRGP-iALS (iALS-p with the DRGP prior) | 90 | – | – | 45 |
| DRGP-LGN-bpr | 72 | – | – | 8 |
| DRGP-LGN-dr | 18 | – | – | 2 |
| DRGP-LGN-hyb | 36 | – | – | – |

Counts are the configurations stored in the result files. The QP family pools the jobs listed above it for each dataset, and its count is their sum. The DRGP prior alone has one configuration per run. Controls are not proposed methods. A dash marks a member that was not run on the dataset.

54 on Coat, 36 on Yahoo! R3, 6 on KuaiRand and 24 on KuaiRec.

A low-rank variant replaces $\hat{Y}$ in $Q$, and in the prior of its offset variant, by a rank-16 low-rank imputation with ridge penalty 5. It has 54 configurations on Coat in the Q-only form and 108 in the offset form.

Three control weightings replace $Q$ in the same learner. The first is $Q$ computed from $\hat{Y}$ alone, with no propensity correction (DR-BPR-yhat; 54 configurations on Coat). The second is $Q$ computed from item popularity, rescaled to the mean of the DR weight (DR-BPR-pop; 54 on Coat and 12 on Yahoo! R3). The third is a fixed random permutation of the values of $Q$ over the matrix (DR-BPR-shuf; 54 on Coat and 12 on Yahoo! R3). The controls test whether the information in $Q$ matters, and they are not proposed methods.

*Appendix B.2. QP-LightGCN*

QP-LightGCN combines the Q weighting with the offset of Section 4.3, so that $f_n = 1$ and the BPR margin contains $\beta(z_{ui} - z_{uj})$. Two offset search spaces are used. DR-BPR-LGN-p has $\beta \in \{0.5, 1\}$, and DR-BPR-LGN-p2 has $\beta \in \{1, 2\}$. The configurations of each offset search space are 108 on

Coat, 24 on Yahoo! R3 (learning rate $3 \cdot 10^{-3}$) and 12 on KuaiRand. On KuaiRec, DR-BPR-LGN-p has 24 configurations (learning rate $3 \cdot 10^{-3}$) and DR-BPR-LGN-p2 has 72 (learning rates $10^{-2}$, $3 \cdot 10^{-3}$ and $10^{-3}$).

The QP-LightGCN family of each dataset pools its jobs. For each split, the job, configuration and epoch with the highest validation score are selected. The family contains DR-BPR-LGN, DR-BPR-LGN-p and DR-BPR-LGN-p2 on Coat, which gives 324 configurations. On Yahoo! R3 it contains DR-BPR-LGN-1, DR-BPR-LGN-p and DR-BPR-LGN-p2, which gives 84 configurations. On KuaiRand and KuaiRec the family contains the same job types, DR-BPR-LGN-1, DR-BPR-LGN-p and DR-BPR-LGN-p2, and it has 30 and 120 configurations respectively.

*Appendix B.3. Ensemble with DRUP*

For each split, component A is the QP-LightGCN family configuration selected for that split, and component B is the DRUP configuration selected for that split. The scores of each component are standardised within each user over the unlogged items of the training log, that is, over the pairs with $O_{ui} = 0$. The standardised score is $\tilde{S}_{ui} = (S_{ui} - \bar{S}_u)/s_u$, with the mean and the standard deviation taken over those pairs. The ensemble score is

$$(1 - w)\, \tilde{S}^A_{ui} + w\, \tilde{S}^B_{ui}, \qquad w \in \{0, 0.1, \ldots, 1\}.$$

The weight $w$ is the first maximiser over this search space of the validation nDCG, and the test score is computed with that $w$. The ensemble stage also stores the rows $w = 0$ (component A), $w = 1$ (DRUP) and $w = 0.5$ as checks. The weight is chosen from validation lists only. The ensemble adds no configuration of its own, so the components carry the configuration counts.

*Appendix B.4. DRGP: a leave-edge-out DR graph prior*

DRGP has two stages. It was run on Coat and KuaiRec only, as single runs without seed repeats. On Coat, 10 of 10 jobs completed, with 222 configurations in total, and on KuaiRec, 5 of 5 jobs completed, with 57 configurations in total (Table B.13).

*Stage A: the prior..* For each pair, define the leave-edge-out 3-hop term of the DR graph. Let $C$ be the degree weights defined above, with the degrees

taken from $\hat{Y}$. Let $\mathrm{sq} = C \circ C \circ W$ and $A_{ui} = \sum_j \mathrm{sq}_{uj} + \sum_v \mathrm{sq}_{vi} - 2\,\mathrm{sq}_{ui}$. Then

$$T_{ui}^{\mathrm{LEO}} = T_{ui} - (W_{ui} - \hat{Y}_{ui})\,(C_{ui}A_{ui} + C_{ui}^3),$$

where $T$ is the walk-corrected three-hop term on $C \circ W$. The term $T_{ui}^{\mathrm{LEO}}$ does not depend on the exposure and outcome of the pair $(u, i)$. The same construction applied to the observed positive graph $O \circ Y$, with base 0, gives a third feature.

The features are $\phi_1 = \operatorname{logit} \hat{Y}$, $\phi_2$ and $\phi_3$ equal to $T^{\mathrm{LEO}}$ of the DR graph with $\alpha \in \{0.5, 0.7\}$, and $\phi_4$ equal to $T^{\mathrm{LEO}}$ of the observed positive graph with $\alpha = 0.5$. Each feature is standardised over the matrix. The prior is

$$\pi_{ui} = a_u + \sum_k \theta_{k,b(u)}\,\phi_{k,ui},$$

where $b(u)$ is a user-activity bucket defined by the quantiles of each user's number of exposures. The coefficients minimise $\sum_{u,i}(\pi_{ui} - W_{ui})^2 + \rho\|\theta\|^2$ with $\rho = 1$. This objective equals the DR squared risk up to a constant for each pair. By the Frisch-Waugh theorem, it is a ridge regression of the row-demeaned $W$ on the row-demeaned features, within each bucket. The imputation inside the features uses penalty $\lambda = 5$ on Coat and $\lambda = 20$ elsewhere. The nuisances of the prior are cross-fitted over ten folds. The prior is the standardised logit $z(\pi)$ of $\pi$, and the number of buckets is 1 or 2 on Coat and 4 on KuaiRec.

*Stage B: the learners..* DRGP-iALS is iALS-p with the prior $z(\pi)$ as its offset. Its search space is that of iALS-p, with 90 configurations on Coat and 45 on KuaiRec. DRGP-LGN-bpr is LightGCN with the BPR margin and the offset $\beta z(\pi)$, with $\beta \in \{0.1, 0.2, 0.5, 1\}$; it has 72 configurations on Coat and 8 on KuaiRec. DRGP-LGN-dr is a LightGCN learner with a self-imputing pointwise doubly robust loss. Writing $\tilde{\pi} = \min\{0.99, \max\{0.01, \pi\}\}$ and $\ell$ for the binary cross-entropy on logits,

$$\frac{1}{mn}\sum_{u,i}\Big[\ell(s_{ui}, \tilde{\pi}_{ui}) + \frac{O_{ui}}{\bar{p}_{ui}}\big(\ell(s_{ui}, Y_{ui}) - \ell(s_{ui}, \tilde{\pi}_{ui})\big)\Big].$$

Its score offset is $\operatorname{logit} \tilde{\pi}$ with weight 1, so the imputed term vanishes at initialisation. It has 18 configurations on Coat and 2 on KuaiRec. DRGP-LGN-hyb adds this loss to the BPR loss with weight $\eta \in \{0.3, 1\}$, and it has 36 configurations on Coat.

*Appendix B.5. Results of the refinement family*

The refinements built on the prior do not improve on the simple prior of the same learner on Coat and KuaiRec (Table B.14). On Coat, the QP family is not different from LightGCN-p (+0.05 [8.29], and the TOST analysis excludes differences larger than 0.010, $p = 0.009$). The largest refinement difference from LightGCN-p is +3.27 [6.94] (DR-BPR-LGN-p2), which is not significant. The Q-only variants are −8.02 [18.85] and −10.79 [19.26] below it. On iALS, DR-iALS is −10.22 [15.96] against iALS-p ($p_{\text{Holm}} = 0.208$). On KuaiRec the QP family is −0.32 [0.50] against LightGCN-p ($p_{\text{Holm}} = 0.211$, within ±0.003). On Yahoo! R3 and KuaiRand-Pure the QP family does exceed LightGCN-p, by +18.20 [5.64] and +4.74 [1.54] (both $p_{\text{Holm}} < 0.001$; Table B.14), so the refinements are not controls there. Every refinement in the study is below DRUP (0.6378) on KuaiRec.

Table B.14: Refinement family against the simple debiased prior of the same learner, and the shuffled and popularity controls against the backbone.

| Dataset | Family | Method | Mean | Cfg | Versus | Diff. $\times 10^3$ [95% CI] | $p_{\text{Holm}}$ | $p_{\text{NB,Holm}}$ | TOST at margin (Coat: exclusion, not equivalence) |
|---|---|---|---|---|---|---|---|---|---|
| Coat | refinement vs simple prior | DR-BPR-LGN | 0.5912 | 108 | LightGCN-p (add prior) | -8.02 [18.85] | 1.000 | 1.000 | does not exclude |
| Coat | refinement vs simple prior | DR-BPR-LGN-1 (Q-only) | 0.5887 | 54 | LightGCN-p (add prior) | -10.79 [19.26] | 1.000 | 1.000 | does not exclude |
| Coat | refinement vs simple prior | DR-BPR-LGN-p | 0.6039 | 108 | LightGCN-p (add prior) | +2.09 [6.48] | 1.000 | 1.000 | excludes $\|d\| > 0.010$ |
| Coat | refinement vs simple prior | DR-BPR-LGN-p2 | 0.6046 | 108 | LightGCN-p (add prior) | +3.27 [6.94] | 1.000 | 1.000 | excludes $\|d\| > 0.010$ |
| Coat | refinement vs simple prior | QP family (pooled) | 0.6015 | 324 | LightGCN-p (add prior) | +0.05 [8.29] | 1.000 | 1.000 | excludes $\|d\| > 0.010$ |
| Coat | refinement vs simple prior (iALS) | DR-iALS | 0.5871 | 48 | iALS-p (add prior) | -10.22 [15.96] | 0.208 | 0.576 | does not exclude |
| Coat | controls vs backbone | control: shuffled Q | 0.5689 | 54 | LightGCN | +0.97 [5.50] | 0.730 | 1.000 | excludes $\|d\| > 0.010$ |
| Coat | controls vs backbone | control: popularity Q | 0.5751 | 54 | LightGCN | +6.91 [6.04] | 0.050 | 1.000 | does not exclude |
| Yahoo | refinement vs simple prior | DR-BPR-LGN-1 (Q-only) | 0.6837 | 36 | LightGCN-p (add prior) | +15.15 [5.84] | <0.001 | 0.031 | no |
| Yahoo | refinement vs simple prior | QP family (pooled) | 0.6866 | 84 | LightGCN-p (add prior) | +18.20 [5.64] | <0.001 | 0.002 | no |
| Yahoo | controls vs backbone | control: shuffled Q | 0.6689 | 12 | LightGCN | -1.01 [3.20] | 0.534 | 0.614 | yes |
| Yahoo | controls vs backbone | control: popularity Q | 0.6550 | 12 | LightGCN | -15.91 [6.06] | <0.001 | 0.343 | no |
| KuaiRand | refinement vs simple prior | QP family (pooled) | 0.4613 | 30 | LightGCN-p (add prior) | +4.74 [1.54] | <0.001 | 0.054 | no |
| KuaiRec | refinement vs simple prior | QP family (pooled) | 0.6285 | 120 | LightGCN-p (add prior) | -0.32 [0.50] | 0.211 | 0.871 | yes |

Holm within each family and dataset; $p_{\text{NB,Holm}}$ applies the Nadeau-Bengio correction of Section 3.5 to the same family. TOST at the sensitivity margin: *yes* means equivalence within ±0.010 (Coat, Yahoo! R3) or ±0.003 (KuaiRand-Pure, KuaiRec) at level 0.05; *no* means that equivalence is not shown. On Coat the entry states only whether differences larger than ±0.010 are excluded, and equivalence is not claimed. Rows are missing where the simple prior has not been run for that dataset.

The leave-edge-out prior of the DRGP family reaches 0.6318 on KuaiRec, which is −5.58 [1.90] against DRUP ($p_{\text{Holm}} < 0.0001$), and 0.5992 on Coat, which is −0.13 [15.11] against QP-LightGCN ($p_{\text{Holm}} = 1.000$). These are single runs without seed repeats (Appendix B.4).

A validated ensemble of QP-LightGCN and DRUP selects one component in most splits and does not add to either: DRUP in four of five KuaiRec splits and QP-LightGCN in nine of ten Coat splits (stored ensemble summary). On KuaiRec the ensemble is −0.18 [0.19] against DRUP ($p_{\text{Holm}} = 0.0664$), and on Coat it is +1.79 [2.39] against QP-LightGCN ($p_{\text{Holm}} = 0.1421$).

The prior alone, without a learner, is a training-free scorer. On KuaiRec the unweighted prior alone reaches 0.6367 and the weighted prior alone 0.6216, and popularity alone reaches 0.5766 (Table 6). On KuaiRand-Pure popularity alone (0.4204) exceeds the weighted prior alone (0.3637) by +56.41 [5.19] ($p_{\text{Holm}} < 0.001$), and the unweighted prior alone (0.3837) exceeds the weighted one by +19.90 [2.45]. On Coat and Yahoo! R3, popularity alone is not significantly different from the weighted prior alone. No paired test of the prior alone against DRUP is in the stored results, so the 0.0011 gap between the unweighted prior alone and DRUP on KuaiRec is a difference of means only.

## Appendix C. Sensitivity and robustness

This appendix collects the sensitivity analyses referred to in Section 5. They use the stored per-split results and the per-epoch records of the search, and they train no model.

### *Appendix C.1. Size of the searched space*

The searched spaces differ in size, from 6 configurations for several KuaiRand-Pure and KuaiRec baselines to 324 for the QP family on Coat. To measure how much this matters, we drew 1,000 random subsets of $k_{\min}$ configurations for each comparison and counted how often its sign changed (Table C.15). Flips are common for comparisons whose effects are near zero. The QP family against LightGCN-p changes sign in 35.3% of the subsets on Coat and 30.9% on KuaiRec. The largest instability is DRUP against DR-JL on KuaiRec. At $k_{\min} = 6$, 78.4% of the subsets flip the sign, and the split-mean difference at matched budget is −14.90, while the full-search-space per-user difference is +4.38 [2.21]. DRUP is training-free but searches 384 configurations on KuaiRec, against 6 for DR-JL, so its advantage over DR-JL depends on the budget granted to each method. The matched-budget values in Table 5 are split means and carry no test.

### *Appendix C.2. Weighting unit, metric and seeds*

The weighting unit can also change a sign. For DR-LightGCN against LightGCN on Coat, the split-mean difference at matched budget ($k = 54$) is +4.11, while the per-user paired difference is −0.65 [30.16]. Every test in this paper uses the per-user unit. Changing the metric changes 3 of

Table C.15: Sensitivity of the primary conclusions to evaluation choices. A check mark means the sign of the difference (and, for Main, significance at $p_{\text{Holm}} < 0.05$) survives the choice; the value is the difference $\times 10^3$ under that choice. Main: per-user difference. Oracle, configuration-oracle and matched-budget columns: split-mean difference, compared in sign with the split-mean validation-selected difference.

| Dataset | Fam. | Comparison | Main (Holm) | Test oracle | Config. oracle | Matched $k$ | $k_{\min}$ | Flip share | Seeds | NB | Alt. metric |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Coat | P1 | DR-LightGCN – LightGCN | × -0.65 | ✓ +1.23 | × -6.63 | ✓ +4.11 | 54 | 0.0% | n/a | × | × |
| Coat | P2 | iALS-p (add prior) – iALS | × +16.82 | ✓ +31.66 | ✓ +21.23 | ✓ +16.12 | 24 | 0.0% | n/a | × | ✓ |
| Coat | P2 | LightGCN-p (add prior) – LightGCN | ✓ +31.98 | ✓ +27.74 | ✓ +28.27 | ✓ +34.12 | 54 | 0.0% | n/a | × | ✓ |
| Coat | P3 | QP family (pooled) – LightGCN-p (add prior) | × +0.05 | × +2.17 | × +1.42 | ✓ -0.36 | 108 | 35.3% | n/a | × | × |
| Coat | P3 | DR-iALS – iALS-p (add prior) | × -10.22 | ✓ -24.87 | ✓ -13.87 | ✓ -11.21 | 48 | 0.0% | n/a | × | ✓ |
| Yahoo | P1 | r-AdjNorm – LightGCN | ✓ +4.10 | ✓ +2.35 | ✓ +0.57 | ✓ +2.87 | 36 | 2.4% | ✓ | × | ✓ |
| Yahoo | P2 | LightGCN-p (add prior) – LightGCN | × -1.43 | ✓ -0.07 | ✓ -1.11 | ✓ -2.16 | 24 | 0.0% | n/a | × | ✓ |
| KuaiRand | P1 | r-AdjNorm – LightGCN | ✓ +1.99 | ✓ +1.76 | ✓ +1.80 | ✓ +1.23 | 6 | 9.4% | ✓ | × | ✓ |
| KuaiRand | P2 | LightGCN-p (add prior) – LightGCN | × -0.81 | ✓ -0.03 | ✓ -0.84 | ✓ -0.79 | 6 | 7.1% | n/a | × | ✓ |
| KuaiRec | P1 | DR-LightGCN – LightGCN | ✓ +13.17 | ✓ +14.69 | ✓ +12.44 | ✓ +12.97 | 6 | 0.0% | n/a | × | × |
| KuaiRec | P2 | LightGCN-p (add prior) – LightGCN | ✓ +21.22 | ✓ +20.02 | ✓ +20.00 | ✓ +20.25 | 24 | 0.0% | n/a | ✓ | ✓ |
| KuaiRec | P2 | iALS-p (add prior) – iALS | ✓ +14.74 | ✓ +14.00 | ✓ +14.00 | ✓ +16.89 | 8 | 0.0% | n/a | ✓ | ✓ |
| KuaiRec | P3 | QP family (pooled) – LightGCN-p (add prior) | × -0.32 | × +0.32 | × +0.46 | ✓ -0.12 | 24 | 30.9% | n/a | × | ✓ |
| KuaiRec | P4 | DRUP – DR-JL | ✓ +4.38 | ✓ +5.48 | ✓ +5.45 | × -14.90 | 6 | 78.4% | n/a | × | ✓ |

Flip share: fraction of 1,000 random subsets of $k_{\min}$ configurations per method whose sign differs from the full search space. Alt. metric: nDCG@10 (Coat, Yahoo), nDCG@20 (KuaiRand), nDCG@50 (KuaiRec), on the stored selections.

the 14 pairwise conclusions in Table C.15: the Coat comparisons of DR-LightGCN with LightGCN and of the QP family with LightGCN-p, and the KuaiRec comparison of DR-LightGCN with LightGCN. The alternative metric is nDCG@10 on Coat and Yahoo! R3, nDCG@20 on KuaiRand-Pure and nDCG@50 on KuaiRec. Of the 14 comparisons in Table C.15, 7 are significant at $p_{\text{Holm}} < 0.05$ in the main run. Over all 14 comparisons, the sign survives the per-split test oracle in 12, the configuration-level oracle in 11, the matched budget in 13 and the alternative metric in 11.

Retraining the selected configurations with seeds 1 to 3 changes two main-run conclusions (the seeds column of Table 5). These comparisons are descriptive, because the two baselines of each dataset were chosen on test data. On Coat, QP-LightGCN against iALS is +20.02 [19.74] in the main run ($p_{\text{Holm}} = 0.0469$, nominal, in a family of eight baselines). Over seeds 1 to 3 the difference is +3.39 [20.70] ($p_{\text{Holm}} = 0.747$, in a family of two), which is not significant. On KuaiRec the main-run deficits of QP-LightGCN against DR-JL and MRDR, −4.56 [2.24] and −4.34 [2.16], shrink over seeds 1 to 3 to −1.03 [1.64] ($p_{\text{Holm}} = 0.2213$) and −1.56 [1.47] ($p_{\text{Holm}} = 0.0760$). On Yahoo! R3 and KuaiRand-Pure the QP-LightGCN family exceeds r-AdjNorm by +14.76 [5.23] and by +2.31 [1.23] over seeds 1 to 3; these seed comparisons are descriptive and carry no significance claim. The seed-to-seed standard deviation of QP-LightGCN on Coat (0.0049, ddof 0) is larger than that of iALS (0.0029), so the main-run Coat result is a single-run result.

*Appendix C.3. Bootstrap ranks*

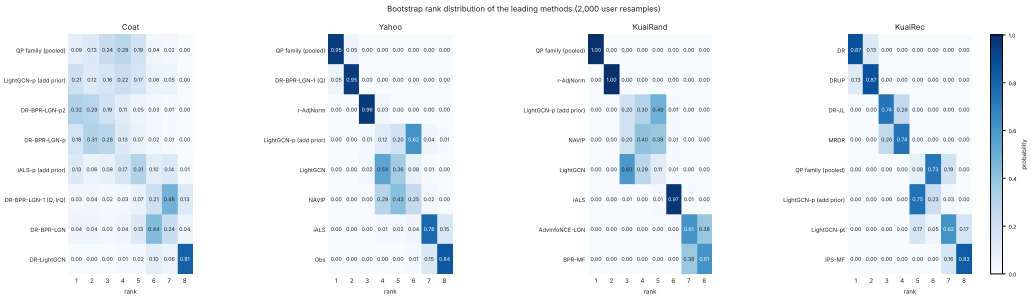



Figure C.1: Bootstrap rank distributions over 2,000 user resamples. For each dataset the rows are the eight methods with the highest mean validation score, excluding control rows, and the columns are the rank by mean test score (1 is best). On Coat the splits are resampled together with the users. Each cell gives the probability of that rank.

The bootstrap ranks (Figure C.1) show how stable the leader is. On Yahoo! R3 the QP family ranks first in 95.0% of user resamples, and on KuaiRand-Pure in 99.8%. On KuaiRec, DR and DRUP have the same mean score (0.6378) and exchange first place, with probabilities 0.873 and 0.127. The bootstrap therefore cannot separate them. On Coat no method is a stable leader. The most frequent first place is DR-BPR-LGN-p2, with probability 0.319, and the QP family ranks first with probability 0.090.

*Appendix C.4. Two checks on the prior inputs*

Coat's shipped propensities were estimated by the dataset authors from the uniform sample, so the prior rows were refitted with the estimator used on the other three datasets (Table C.16). The estimated-minus-shipped difference is +2.63 [5.27] for iALS-p and +1.64 [10.30] for LightGCN-p, and the imputation alone changes by −1.14 [7.26]. None is significant. The gain over the no-prior reference remains as a point estimate: +19.44 [18.66] for iALS-p ($p_{\text{Holm}} = 0.330$, not significant) and +33.62 [16.72] for LightGCN-p. The estimated-propensity run of LightGCN was not carried out, and the other propensity-based Coat rows were not refitted.

The stored iALS search space selects its largest value of the confidence weight $\alpha$ (30) in all 10 splits, and the wider search space of Table C.17 selects its largest value (1000) in four splits. The stored search space is therefore bounded at its edge, and the wider one is not exhausted. Over

Table C.16: Coat with estimated propensities versus the shipped propensities.

| Method | Propensities | nDCG (mean ± sd) | Δ vs no-prior $\times 10^3$ [CI] | $p_{\text{Holm}}$ | Estimated − shipped $\times 10^3$ [CI] | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|---|
| iALS-p | shipped | 0.5984 ± 0.0128 | +16.82 [18.04] | 0.380 | – | – |
| iALS-p | estimated | 0.6009 ± 0.0120 | +19.44 [18.66] | 0.330 | +2.63 [5.27] | 1.000 |
| LightGCN-p | shipped | 0.6020 ± 0.0156 | +31.98 [20.65] | 0.023 | – | – |
| LightGCN-p | estimated | 0.6025 ± 0.0154 | +33.62 [16.72] | <0.001 | +1.64 [10.30] | 1.000 |
| iALS (search B), sanity check | shipped | 0.5869 ± 0.0124 | +7.39 [7.52] | 0.380 | – | – |
| iALS (search B), sanity check | estimated | 0.5869 ± 0.0124 | +7.39 [7.52] | 0.380 | 0 (identical) | 1 |
| imputation alone (add) | shipped | 0.5585 ± 0.0105 | – | – | – | – |
| imputation alone (add) | estimated | 0.5573 ± 0.0097 | – | – | -1.14 [7.26] | 1.000 |
| iALS (no prior) | shipped | 0.5812 ± 0.0107 | – | – | – | – |
| LightGCN (no prior) | shipped | 0.5683 ± 0.0139 | – | – | – | – |

The no-prior reference of each learner is the stored run. Plain iALS and LightGCN do not read the propensities (only NAVIP does), so the shipped no-prior run is the reference for the estimated rows. LightGCN with estimated propensities was not run. The iALS (search B) row is a sanity check: plain iALS does not read the propensities, so its difference is zero by construction. Only the prior rows and the imputation alone were refitted. $p_{\text{Holm}}$ over the tests of this table.

the stored iALS, the wider search space gives +12.92 [8.01] ($p_{\text{Holm}} = 0.003$), against +16.82 [18.04] for the offset model over the same stored iALS. The wider search space is not significantly different from iALS-p (−3.90 [15.99], $p_{\text{Holm}} = 0.631$). The two differences cannot be separated: the wider search space has 32 configurations against 90 for iALS-p.

Table C.17: Coat iALS with the wider search space against the stored iALS and iALS-p.

| Model | Search space | Cfg | nDCG (mean ± sd) | Δ vs iALS $\times 10^3$ [CI] | $p_{\text{Holm}}$ | Δ vs iALS-p $\times 10^3$ [CI] | $p_{\text{Holm}}$ | Selected values (number of splits) |
|---|---|---|---|---|---|---|---|---|
| iALS (search A, stored) | alpha {3,10,30} x reg {1,10,100,1000} x dim {32,64} | 24 | 0.5812 ± 0.0107 | ref. | – | – | – | alpha 30:10; reg 100:10 |
| iALS (search B) | alpha {10,30,100} x reg {10,100,300} x dim {32,64} | 18 | 0.5869 ± 0.0124 | – | – | – | – | alpha 30:1, 100:9; reg 100:4, 300:6 |
| iALS (search C) | alpha {30,100,300,1000} x reg {10,100,300,1000} x dim {32,64} | 32 | 0.5922 ± 0.0150 | +12.92 [8.01] | 0.003 | -3.90 [15.99] | 0.631 | alpha 30:1, 100:4, 300:1, 1000:4; reg 100:2, 300:3, 1000:5 |
| iALS-p (add, stored search) | alpha {10,30,100} x reg {10,100,300} x pbeta {0.1,0.2,0.3,0.5,1} x dim {32,64} | 90 | 0.5984 ± 0.0128 | – | – | – | – | alpha 30:2, 100:8; reg 100:4, 300:6 |

$\Delta$ = per-user paired difference (mean over the 10 splits), $\times 10^3$, [half-width of the 95% CI]. $p_{\text{Holm}}$ over the two comparisons of search space C. Selected values are counted over the 10 splits. Search space C was widened after search space B and is not exhausted (alpha 1000 is chosen in 4 splits).

## Appendix D. Walk correction: synthetic, semi-synthetic and training-free evidence

This appendix gives the evidence on the walk correction that is not based on the real logs, and the training-free operators on KuaiRec. The values are read from the stored outputs of the synthetic and semi-synthetic studies and of the training-free runs.

### *Appendix D.1. Synthetic MNAR study*

On a synthetic MNAR model with independent exposure, the walk correction reduces the relative |bias| of the DR adjacency from 1.756 to 0.019 at

$K = 3$ and from 8.350 to 0.039 at $K = 5$ (stored synthetic-study outputs). This is a bias result on a model in which the true propensities and outcomes are known. It does not measure ranking on real logs.

*Appendix D.2. Semi-synthetic KuaiRec study*

The semi-synthetic study builds its logs from the dense KuaiRec setting, with 30 replications, $\beta = 1$, true propensities and fixed degree weights. The relative bias of the three-hop term is 1.95 for DR and 0.88 for DRUP, and 3.46 against 0.32 at $K = 5$. With estimated propensities the same comparison gives 9.47 against 5.66. In the three cells with degree weights computed from the imputation and true propensities, the walk correction changes the relative bias from 0.67 to 0.66 (independent nuisance fit), from 0.72 to 0.75 (cross-fitted) and from 0.53 to 0.32 (20% split). Table D.19 gives all cells.

The same study scores the candidates by nDCG@20 against the true labels on unexposed items (Table D.18). DRUP is higher than DR in seven of the nine cells. In the two cells with imputed degree weights and an independent or a 20% split nuisance fit, DR is higher (0.3769 against 0.3697, and 0.2742 against 0.2723). The differences of means are not paired tests, because their standard errors are not stored. These are results for a known-label world with a known exposure mechanism, and they are not evidence about the real logs, where the correction did not raise nDCG (Section 5.4).

Table D.18: Semi-synthetic KuaiRec study: nDCG@20 on unexposed candidates against the true labels. Mean over 30 replications with the standard error in parentheses.

| Nuisance fit | Propensity | Degree weights | IPS | IPS+WC | DR | DRUP | Δ DRUP−DR |
|---|---|---|---|---|---|---|---|
| independent | true | fixed | 0.0967 (0.0011) | 0.0967 (0.0011) | 0.0973 (0.0015) | 0.1071 (0.0012) | +0.0099 |
| independent | true | imputed | 0.1202 (0.0018) | 0.1202 (0.0018) | 0.3769 (0.0011) | 0.3697 (0.0011) | -0.0072 |
| independent | estimated | fixed | 0.0569 (0.0005) | 0.0569 (0.0005) | 0.0294 (0.0004) | 0.0379 (0.0003) | +0.0085 |
| cross-fitted | true | fixed | 0.0967 (0.0011) | 0.0967 (0.0011) | 0.0510 (0.0011) | 0.0683 (0.0010) | +0.0173 |
| cross-fitted | true | imputed | 0.1915 (0.0018) | 0.1915 (0.0018) | 0.3203 (0.0009) | 0.3218 (0.0009) | +0.0015 |
| cross-fitted | estimated | fixed | 0.0559 (0.0004) | 0.0559 (0.0004) | 0.0206 (0.0003) | 0.0303 (0.0004) | +0.0097 |
| 20% split | true | fixed | 0.0926 (0.0012) | 0.0913 (0.0011) | 0.0303 (0.0009) | 0.0479 (0.0013) | +0.0176 |
| 20% split | true | imputed | 0.1545 (0.0012) | 0.1500 (0.0012) | 0.2742 (0.0029) | 0.2723 (0.0027) | -0.0019 |
| 20% split | estimated | fixed | 0.0657 (0.0007) | 0.0711 (0.0007) | 0.0198 (0.0004) | 0.0293 (0.0004) | +0.0096 |

The last column is the difference of the two means, which is not a paired test: the standard error of the difference is not stored. Higher values are better. Nuisance fit: *independent* uses a log independent of the one scored, *cross-fitted* uses folds of the same log, and *20% split* uses a 20% sample split. The degree weights are *fixed* (the true-degree normalisation) or *imputed* from the imputation.

Table D.19: Semi-synthetic KuaiRec study: relative bias of the three-hop term, averaged over the draws. The last row is the five-hop term.

| Nuisance fit | Propensity | Degree weights | IPS | IPS+WC | DR | DRUP |
|---|---|---|---|---|---|---|
| independent | true | fixed | 0.56 | 0.27 | 1.95 | 0.88 |
| independent | true | imputed | 0.66 | 0.65 | 0.67 | 0.66 |
| independent | estimated | fixed | 6.37 | 4.80 | 9.47 | 5.66 |
| cross-fitted | true | fixed | 0.56 | 0.27 | 2.29 | 1.10 |
| cross-fitted | true | imputed | 0.71 | 0.74 | 0.72 | 0.75 |
| cross-fitted | estimated | fixed | 6.26 | 4.70 | 10.88 | 6.47 |
| 20% split | true | fixed | 0.81 | 0.55 | 2.09 | 1.02 |
| 20% split | true | imputed | 0.47 | 0.31 | 0.53 | 0.32 |
| 20% split | estimated | fixed | 9.47 | 7.90 | 11.78 | 8.19 |
| independent (K=5) | true | fixed | – | – | 3.46 | 0.32 |

Relative bias: mean absolute bias of the averaged estimate over the entries of the three-hop term, divided by the mean absolute true three-hop term. IPS and IPS+WC (the walk-corrected IPS estimator) are shown for reference. The column definitions are those of Table D.18.

### *Appendix D.3. Training-free operators on KuaiRec*

Among the training-free operators, the best KuaiRec mean goes to doubly robust propagation, not to the walk correction. DRUP reaches 0.6378, the DR adjacency on which it is built also reaches 0.6378, and a five-hop DR adjacency reaches 0.6379 (stored training-free runs). No trained model in Table 4 exceeds 0.6378 on KuaiRec: the best is DR-JL at 0.6332, and the QP family reaches 0.6285. On Coat, DRUP is weak at 0.5467, and on Yahoo! R3 the DRUP and DR rows coincide at 0.6567. In the bootstrap, DR ranks first in 87.3% of resamples on KuaiRec and DRUP in 12.7% (Figure C.1), which reflects the tie in their means. The unweighted prior alone (0.6367) is the one training-free result close to DRUP, and it is reported in Table 6. The reason the walk correction does not change the ranking on these logs is not identified by our runs.

## Appendix E. Search spaces, variants and reconciliation

This appendix lists the search spaces of the trained baselines and the AdvInfoNCE re-implementation, the prior variants, the reconciliation of the stored results, and the protocol of each related study (Section 2).

*Appendix E.1. Trained baselines and AdvInfoNCE*

*Coat..* The search space is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}, 10^{-3}\}$ and width in $\{32, 64\}$. For LightGCN, LightGCN-pt, r-AdjNorm, NAVIP, DR-LightGCN and SimGCL the depth ranges over $\{1, 2, 3\}$. Epochs are limited to 100, the batch size is 256 and the patience is 5. Configurations: LightGCN 54, LightGCN-pt 54, NAVIP 54, DR-LightGCN 54, r-AdjNorm 108 (with $r \in \{0.3, 0.7\}$), SimGCL 108 (with weight $\lambda_{\text{cl}} \in \{0.1, 0.5\}$), MACR 36 (with $c \in \{0.1, 0.3\}$), PDA 36 (with $\gamma \in \{0.1, 0.2\}$), MF, IPS-MF, DR-MF, BPR-MF, DR-JL and MRDR 18 each, iALS 24 (with $\alpha \in \{3, 10, 30\}$, regularisation in $\{1, 10, 100, 1000\}$ and width in $\{32, 64\}$) and iALS search space B 18 (with $\alpha \in \{10, 30, 100\}$ and regularisation in $\{10, 100, 300\}$). The iALS runs of the prior family use an epoch limit of 30 on Coat, and the stored iALS run uses 100. The epoch limits are therefore not the same across the iALS rows.

*Yahoo! R3..* The search space is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$, weight decay in $\{10^{-5}, 10^{-4}\}$ (with $10^{-3}$ added for the matrix models), width 64 and depth 2. For LightGCN, NAVIP and r-AdjNorm the width ranges over $\{64, 128\}$ and the depth over $\{1, 2, 3\}$. Epochs are limited to 60, the batch size is 1,024 and the patience is 5. Configurations: LightGCN 36, NAVIP 36, r-AdjNorm 72, LightGCN-pt 6, DR-LightGCN 6, SimGCL 12, MACR 18, PDA 18, iALS 12 (with the iALS search space above and width 64), MF, IPS-MF, DR-MF, BPR-MF, DR-JL and MRDR 9 each.

*KuaiRand-Pure..* The search space is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$ and weight decay in $\{10^{-5}, 10^{-4}\}$, with width 64 and depth 2 for every model. Epochs are limited to 40, the batch size is 8,192 and the patience is 3. Configurations: LightGCN 6, LightGCN-pt 6, NAVIP 6, DR-JL 6, MF 6, DR-MF 6, BPR-MF 6, r-AdjNorm 12, SimGCL 12, MACR 12 and iALS 12. MRDR, IPS-MF, PDA and DR-LightGCN were not run on this dataset.

*KuaiRec..* The search space is learning rate in $\{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}\}$ and weight decay in $\{10^{-5}, 10^{-4}\}$. For LightGCN and LightGCN-pt the width ranges over $\{64, 128\}$ and the depth over $\{1, 2, 3\}$, and the other models use width 64 and depth 2. Epochs are limited to 30, the batch size is 32,768, the patience is 3, and each epoch is capped at 2,000,000 pairs. Configurations: LightGCN 36, LightGCN-pt 36, r-AdjNorm 12, NAVIP 6, DR-LightGCN 6, SimGCL 12, MACR 12, iALS 12, MF 6, IPS-MF 6, DR-MF 6, BPR-MF 6,

DR-JL 6 and MRDR 6. PDA (12 configurations) and two pointwise negative-sampling variants, LightGCN-pn and DR-LightGCN-pn (24 configurations each), were also run on KuaiRec and are not in the main comparison.

Across datasets, the graph models were given depth search spaces $\{1, 2, 3\}$ on Coat, Yahoo! R3 and KuaiRec, and only depth 2 on KuaiRand. Budgets are therefore unequal, and the counts make the difference visible. The selection rule of Section 3.3 is applied to each search space as it stands, and the sensitivity of the conclusions to the size of the search space is measured in the results.

*Appendix E.2. AdvInfoNCE re-implementation*

AdvInfoNCE (Zhang et al., 2023) is an adversarial contrastive loss. We re-implemented it on the LightGCN encoder. Our numbers are for this re-implementation and not for the original method, and the official code was not run. The main simplifications are these.

- The score is the cosine similarity of the propagated embeddings divided by a temperature $\tau$. Each positive is contrasted with 64 sampled negatives.
- The adversary is a pair of embedding tables (users and items, dimension 64) trained with Adam to maximise the contrastive loss. Its learning rate is $10^{-2}$ on Coat, $10^{-3}$ on Yahoo! R3 and KuaiRand, and $5 \cdot 10^{-4}$ on KuaiRec. Only this embedding-based adversary is implemented.
- The adversary budget of the original formulation is replaced by a number of adversarial rounds $\eta \in \{4, 12\}$. A round runs after each epoch from the second epoch on and lasts at least one pass over the positives and 100 Adam steps. The adversary's KL divergence is logged but not projected onto a ball.

The search spaces differ in size, and the budgets are unequal. Coat has 216 configurations: three learning rates, three weight decays, widths $\{32, 64\}$, depths $\{1, 2, 3\}$, $\tau \in \{0.2, 0.75\}$ and $\eta \in \{4, 12\}$. Yahoo! R3 has 16 configurations with $\tau \in \{0.1, 0.28\}$, $\eta \in \{4, 12\}$, learning rates $\{3 \cdot 10^{-3}, 10^{-3}\}$, weight decays $\{10^{-5}, 10^{-4}\}$, width 128 and depth 2. KuaiRand has 8 configurations with $\tau \in \{0.1, 0.28\}$, learning rates $\{10^{-2}, 3 \cdot 10^{-3}\}$, weight decay $10^{-4}$, width 64 and depth 2. KuaiRec has 8 configurations with $\tau \in \{0.2, 2.0\}$, learning rates $\{10^{-2}, 3 \cdot 10^{-3}\}$, weight decay $10^{-5}$, width 64 and depth 2. KuaiRand

and KuaiRec both use $\eta \in \{4, 12\}$. On Coat, AdvInfoNCE thus has 216 configurations against 54 for LightGCN and 108 for r-AdjNorm. A larger search space gives a method more chances to fit noise in the validation data, so the Coat search space favours AdvInfoNCE and the smaller search spaces work against it. We did not match budgets for this baseline. The only wider re-tuning in this study is the Coat iALS search space (Table C.17).

*Appendix E.3. Prior variants*

The variants below change the additive imputation or the way it enters the score. Each is defined in the code of the repository. In every variant, the prior is the standardised logit of the variant's imputation, except for `pop`, which replaces the offset by a popularity score.

**add.** The imputation of Section 4.3, fitted with weights $O/\bar{p}$.

**add_noips.** The same ridge fit with weights $O$ in place of $O/\bar{p}$, so that the fit uses the exposed pairs without inverse-propensity weights.

**pop.** The imputation, and therefore $Q$, is unchanged. The offset is replaced by the standardised logarithm of the number of positives of each item in the log, $\log(1 + \sum_v O_{vi} Y_{vi})$, broadcast to every user.

**shuf_item.** The item biases $b_i$ are permuted across items by a fixed permutation (seed 0). The parameters $\mu$ and $b_u$ are kept, and the imputation and the prior are recomputed from the permuted biases.

**user_only.** The item biases are set to zero, so every item has the same score within a user. The ranking within a user is then decided by tie-breaking.

In the learners of this analysis, only the prior enters the score. The variant's $Q$ is used only by the Q-weighted methods, and the prior-anatomy runs do not use them.

The prior-anatomy runs are the following. On Coat, iALS-p and LightGCN-p are run over the ten splits with the variants add, add_noips, pop and shuf_item. On KuaiRec, iALS-p is run over the five splits with the same four variants on the eight-configuration search space. The prior alone is run for all five variants on all four datasets, with the floors listed in Section 4.3. The floor of the Coat LightGCN-p runs is 0.05, and that of the Coat iALS-p runs is 0.1.

*Appendix E.4. Reconciliation of the stored results*

Before the analysis, the selection of each (file, method) pair was recomputed from its stored per-epoch records. The recomputation applies the training-time rule of Section 3.3, including the patience value recorded with the records, and its mean test score is compared with the stored mean. Of the 154 (file, method) pairs in the reconciliation, 151 agree within 0.002 in nDCG. The three that do not agree are DR-LightGCN on KuaiRec (recomputed mean higher by 0.0033), DR-LightGCN on Yahoo! R3 (higher by 0.0058) and MACR on Yahoo! R3 (higher by 0.0060). All three come from two earlier result files that do not record the patience value. The stored selections are reproduced only under patience values of 3 (KuaiRec) and 5 (Yahoo! R3), which the files do not record. The KuaiRec DR-LightGCN value enters the P1 comparison of Table 5, and the Yahoo! R3 DR-LightGCN and MACR values appear in Table 4. Because the selections cannot be recomputed from the records as stored, we report the stored values. Every sensitivity number in the results is reported relative to the reconciled selection.

*Appendix E.5. Positioning of related studies*

## References


Anelli, V.W., Malitesta, D., Pomo, C., Bellogin, A., Di Sciascio, E., Di Noia, T., 2023. Challenging the myth of graph collaborative filtering: a reasoned and reproducibility-driven analysis, in: Proceedings of the 17th ACM Conference on Recommender Systems, ACM. pp. 350–361. doi:10.1145/3604915.3609489.

Bonner, S., Vasile, F., 2018. Causal embeddings for recommendation, in: Proceedings of the 12th ACM Conference on Recommender Systems, pp. 104–112. doi:10.1145/3240323.3240360.

Chen, C., Lin, H., Xu, Y., 2026. Towards reliable negative sampling for recommendation with implicit feedback via in-community popularity. arXiv:2602.18759. preprint.

Chen, J., Dong, H., Qiu, Y., He, X., Xin, X., Chen, L., Lin, G., Yang, K., 2021. AutoDebias: Learning to debias for recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 21–30. doi:10.1145/3404835.3462919.

Table E.20: Positioning of related studies by their protocol. Entries follow the protocols as described in the cited papers; "not stated" marks a point that the description does not settle. Uniform data are the uniformly collected or randomised samples of the benchmark.

| Study | Graph backbone | Propensity or imputation | Uniform data in training | Uniform data for validation or selection | Unbiased test benchmarks |
|---|---|---|---|---|---|
| AutoDebias (Chen et al., 2021) | no (matrix factorisation) | learned weights and imputation | yes, small sample | yes, small sample | Coat, Yahoo! R3 |
| CausE (Bonner and Vasile, 2018) | no (matrix factorisation) | none | yes, randomised sample | no, validation from logged policy | MovieLens, Netflix |
| CDR (Song et al., 2023) | no (matrix factorisation) | doubly robust family | no | yes, part of unbiased data | Coat, Yahoo! R3, KuaiRand |
| DCE (Kweon and Yu, 2024) | no (MF or NCF) | doubly robust loss | no | no, held out from biased training data | Coat, Yahoo! R3, KuaiRec |
| BISER (Lee et al., 2022) | no | self-inverse propensity | no | no, held out from biased training data | Coat, Yahoo! R3 |
| ConAlign (Zhang et al., 2026) | no (dual tower) | MF-IPS and MF-DR baselines | yes | yes | Coat, Yahoo! R3, KuaiRand-Pure |
| NAVIP (Kim et al., 2022) | yes | inverse propensity on edges | no | pseudo-unbiased popularity sample | Yahoo! R3 and pseudo-unbiased sets |
| AdvDrop (Zhang et al., 2024) | yes | IPS and DR baselines only | no, for its own method | not stated | Coat, Yahoo!, KuaiRec |
| PPD (Islam et al., 2026b) | yes, post hoc | none | no | yes on Coat and KuaiRec; biased sample on Yahoo! R3 | Coat, Yahoo! R3, KuaiRec |
| DPAA (Islam et al., 2026a) | yes | none | no | validation present, construction not confirmed | Coat, Yahoo! R3, KuaiRec |
| DR-GNN (Wang et al., 2024) | yes | none | not stated | not stated | Coat, Yahoo |
| This study | yes | IPS-fitted imputation and propensity | no, except Coat shipped propensities | yes | Coat, Yahoo! R3, KuaiRand-Pure, KuaiRec |

Chen, J., Dong, H., Wang, X., Feng, F., Wang, M., He, X., 2023. Bias and debias in recommender system: A survey and future directions. ACM Transactions on Information Systems 41, 1–39. doi:10.1145/3564284.

Chen, J., Wu, J., Chen, J., Xin, X., Li, Y., He, X., 2024. How graph convolutions amplify popularity bias for recommendation? Frontiers of Computer Science 18. doi:10.1007/s11704-023-2655-2.

Choi, J., Hong, S., Park, N., Cho, S.B., 2023. Blurring-sharpening process models for collaborative filtering, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 1096–1106. doi:10.1145/3539618.3591645.

Chuang, C.Y., Robinson, J., Lin, Y.C., Torralba, A., Jegelka, S., 2020. Debiased contrastive learning, in: Advances in Neural Information Processing Systems 33 (NeurIPS 2020). URL: https://papers.nips.cc/paper/2020/hash/63c3ddcc7b23daa1e42dc41f9a44a873-Abstract.html, arXiv:2007.00224.

Dai, Q., Li, H., Wu, P., Dong, Z., Zhou, X.H., Zhang, R., Zhang, R., Sun, J., 2022. A generalized doubly robust learning framework for debiasing post-click conversion rate prediction, in: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, ACM. pp. 252–262. doi:10.1145/3534678.3539270.

Ding, J., Quan, Y., Yao, Q., Li, Y., Jin, D., 2020. Simplify and robustify negative sampling for implicit collaborative filtering, in: Advances in Neural Information Processing Systems 33 (NeurIPS 2020). URL: https://papers.nips.cc/paper/2020/hash/0c7119e3a6a2209da6a5b90e5b5b75bd-Abstract.html, arXiv:2009.03376.

Dodge, J., Gururangan, S., Card, D., Schwartz, R., Smith, N.A., 2019. Show your work: Improved reporting of experimental results, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 2185–2194. doi:10.18653/v1/D19-1224.

Ferrari Dacrema, M., Cremonesi, P., Jannach, D., 2019. Are we really making much progress? A worrying analysis of recent neural recommendation

approaches, in: Proceedings of the 13th ACM Conference on Recommender Systems, ACM. pp. 101–109. doi:10.1145/3298689.3347058.

Gao, C., Li, S., Lei, W., Chen, J., Li, B., Jiang, P., He, X., Mao, J., Chua, T.S., 2022a. KuaiRec: A fully-observed dataset and insights for evaluating recommender systems [data set], in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM), pp. 540–550. doi:10.1145/3511808.3557220.

Gao, C., Li, S., Zhang, Y., Chen, J., Li, B., Lei, W., Jiang, P., He, X., 2022b. KuaiRand: An unbiased sequential recommendation dataset with randomly exposed videos [data set], in: Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM), pp. 3953–3957. doi:10.1145/3511808.3557624.

Gao, C., Zheng, Y., Wang, W., Feng, F., He, X., Li, Y., 2024. Causal inference in recommender systems: A survey and future directions. ACM Transactions on Information Systems 42, 1–32. doi:10.1145/3639048.

Guo, S., Zou, L., Liu, Y., Ye, W., Cheng, S., Wang, S., Chen, H., Yin, D., Chang, Y., 2021. Enhanced doubly robust learning for debiasing post-click conversion rate estimation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 275–284. doi:10.1145/3404835.3462917.

He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., Wang, M., 2020. LightGCN: Simplifying and powering graph convolution network for recommendation, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 639–648.

Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6, 65–70. URL: https://www.jstor.org/stable/4615733.

Hu, Y., Koren, Y., Volinsky, C., 2008. Collaborative filtering for implicit feedback datasets, in: 2008 Eighth IEEE International Conference on Data Mining (ICDM), pp. 263–272. doi:10.1109/ICDM.2008.22.

Islam, M.A., Faruk, A.S., Medya, S., Zheleva, E., 2026a. Debiasing message passing to mitigate popularity bias in GNN-based collaborative filtering. arXiv:2605.11145. preprint.

Islam, M.A., Zheleva, E., Wang, R., 2026b. Post-hoc popularity bias correction in GNN-based collaborative filtering, in: Proceedings of the ACM Web Conference 2026 (WWW '26), Dubai, United Arab Emirates. doi:10.1145/3774904.3792670, arXiv:2510.12959.

Kim, M., Oh, J., Do, J., Lee, S., 2022. Debiasing neighbor aggregation for graph neural network in recommender systems, in: Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM), pp. 4128–4132. doi:10.1145/3511808.3557576.

Krichene, W., Rendle, S., 2020. On sampled metrics for item recommendation, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, ACM. pp. 1748–1757. doi:10.1145/3394486.3403226.

Kweon, W., Yu, H., 2024. Doubly calibrated estimator for recommendation on data missing not at random, in: Proceedings of the ACM Web Conference 2024 (WWW '24), Singapore. doi:10.1145/3589334.3645417, arXiv:2403.00817.

Lee, J.w., Park, S., Lee, J., Lee, J., 2022. Bilateral self-unbiased learning from biased implicit feedback, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '22), Madrid, Spain. arXiv:2207.12660.

Li, H., Zheng, C., Wu, P., 2023. StableDR: Stabilized doubly robust learning for recommendation on data missing not at random, in: International Conference on Learning Representations (ICLR).

Liang, D., Charlin, L., McInerney, J., Blei, D.M., 2016. Modeling user exposure in recommendation, in: Proceedings of the 25th International Conference on World Wide Web, pp. 951–961. doi:10.1145/2872427.2883090.

Liu, D., Cheng, P., Dong, Z., He, X., Pan, W., Ming, Z., 2020. A general knowledge distillation framework for counterfactual recommendation via uniform data, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20), pp. 831–840. doi:10.1145/3397271.3401083. full text not read; bibliographic data from the ACM reference format quoted by a secondary source. UNVERIFIED against the proceedings.

Mao, K., Zhu, J., Xiao, X., Lu, B., Wang, Z., He, X., 2021. UltraGCN: Ultra simplification of graph convolutional networks for recommendation, in: Proceedings of the 30th ACM International Conference on Information and Knowledge Management, pp. 1253–1262. doi:10.1145/3459637.3482291.

Marlin, B.M., Zemel, R.S., 2009. Collaborative prediction and ranking with non-random missing data [data set], in: Proceedings of the Third ACM Conference on Recommender Systems (RecSys), pp. 5–12. doi:10.1145/1639714.1639717.

Menon, A.K., Jayasumana, S., Rawat, A.S., Jain, H., Veit, A., Kumar, S., 2021. Long-tail learning via logit adjustment, in: International Conference on Learning Representations (ICLR). arXiv:2007.07314.

Nadeau, C., Bengio, Y., 2003. Inference for the generalization error. Machine Learning 52, 239–281. doi:10.1023/a:1024068626366.

Que, Y., Zhang, Y., Zhao, X., Ma, C., 2025. Causality-aware graph aggregation weight estimator for popularity debiasing in top-k recommendation, in: Proceedings of the 34th ACM International Conference on Information and Knowledge Management, ACM. pp. 2471–2481. doi:10.1145/3746252.3761155.

Raja, R., Vats, A., 2025. Counterfactual risk minimization with IPS-weighted BPR and self-normalized evaluation in recommender systems, in: Causality, Counterfactuals and Sequential Decision-Making Workshop at RecSys 2025. arXiv:2509.00333. workshop paper.

Rendle, S., Freudenthaler, C., Gantner, Z., Schmidt-Thieme, L., 2009. BPR: Bayesian personalized ranking from implicit feedback, in: Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI), pp. 452–461. arXiv:1205.2618.

Rendle, S., Krichene, W., Zhang, L., Koren, Y., 2022. Revisiting the performance of iALS on item recommendation benchmarks, in: Proceedings of the 16th ACM Conference on Recommender Systems, ACM. pp. 427–435. doi:10.1145/3523227.3548486, arXiv:2110.14037.

Rendle, S., Zhang, L., Koren, Y., 2019. On the difficulty of evaluating baselines: A study on recommender systems. arXiv:1905.01395. preprint.

Saito, Y., 2020. Unbiased pairwise learning from biased implicit feedback, in: Proceedings of the 2020 ACM SIGIR on International Conference on Theory of Information Retrieval (ICTIR), ACM. pp. 5–12. doi:10.1145/3409256.3409812.

Saito, Y., Yaginuma, S., Nishino, Y., Sakata, H., Nakata, K., 2020. Unbiased recommender learning from missing-not-at-random implicit feedback, in: Proceedings of the 13th International Conference on Web Search and Data Mining (WSDM), pp. 501–509.

Schnabel, T., Swaminathan, A., Singh, A., Chandak, N., Joachims, T., 2016. Recommendations as treatments: Debiasing learning and evaluation [data set], in: Proceedings of the 33rd International Conference on Machine Learning (ICML), pp. 1670–1679.

Schuirmann, D.J., 1987. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. Journal of Pharmacokinetics and Biopharmaceutics 15, 657–680. doi:10.1007/BF01068419.

Shen, Y., Wu, Y., Zhang, Y., Shan, C., Zhang, J., Letaief, K.B., Li, D., 2021. How powerful is graph convolution for recommendation?, in: Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM), pp. 1619–1629.

Song, Z., Chen, J., Zhou, S., Shi, Q., Feng, Y., Chen, C., Wang, C., 2023. CDR: Conservative doubly robust learning for debiased recommendation, in: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23), Birmingham, United Kingdom. doi:10.1145/3583780.3614805, arXiv:2308.08461.

Steck, H., 2010. Training and testing of recommender systems on data missing not at random, in: Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM. pp. 713–722. doi:10.1145/1835804.1835895.

Steck, H., 2019. Embarrassingly shallow autoencoders for sparse data, in: The World Wide Web Conference, ACM. pp. 3251–3257. doi:10.1145/3308558.3313710.

Swaminathan, A., Joachims, T., 2015. The self-normalized estimator for counterfactual learning, in: Advances in Neural Information Processing Systems, pp. 3231–3239.

Wang, B., Chen, J., Li, C., Zhou, S., Shi, Q., Gao, Y., Feng, Y., Chen, C., Wang, C., 2024. Distributionally robust graph-based recommendation system, in: Proceedings of the ACM Web Conference 2024 (WWW '24), Singapore. doi:10.1145/3589334.3645598, arXiv:2402.12994.

Wang, X., He, X., Wang, M., Feng, F., Chua, T.S., 2019a. Neural graph collaborative filtering, in: Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 165–174. doi:10.1145/3331184.3331267.

Wang, X., Zhang, R., Sun, Y., Qi, J., 2019b. Doubly robust joint learning for recommendation on data missing not at random, in: Proceedings of the 36th International Conference on Machine Learning (ICML), pp. 6638–6647.

Wang, Z., Chen, X., Wen, R., Huang, S.L., Kuruoglu, E.E., Zheng, Y., 2020. Information theoretic counterfactual learning from missing-not-at-random feedback, in: Advances in Neural Information Processing Systems.

Wei, T., Feng, F., Chen, J., Wu, Z., Yi, J., He, X., 2021. Model-agnostic counterfactual reasoning for eliminating popularity bias in recommender system, in: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 1791–1800. doi:10.1145/3447548.3467289.

Wu, J., Wang, X., Feng, F., He, X., Chen, L., Lian, J., Xie, X., 2021. Self-supervised graph learning for recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 726–735. doi:10.1145/3404835.3462862.

Wu, S., Sun, F., Zhang, W., Xie, X., Cui, B., 2023. Graph neural networks in recommender systems: A survey. ACM Computing Surveys 55, 1–37. doi:10.1145/3535101.

Yin, M., Hu, C., Liu, B., Gong, N.Z., Ye, Y., Li, X., 2026. Correct and weight: A simple yet effective loss for implicit feedback recommendation. arXiv:2601.04291. preprint.

Yu, J., Yin, H., Xia, X., Chen, T., Cui, L., Nguyen, Q.V.H., 2022. Are graph augmentations necessary?: Simple graph contrastive learning for recommendation, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 1294–1303. doi:10.1145/3477495.3531937.

Zhang, A., Ma, W., Wang, X., Chua, T.S., 2022. Incorporating bias-aware margins into contrastive loss for collaborative filtering, in: Advances in Neural Information Processing Systems 35 (NeurIPS 2022). URL: https://proceedings.nips.cc/paper_files/paper/2022/file/334da4cbb76302f37bd2e9d86f558869-Paper-Conference.pdf, arXiv:2210.11054.

Zhang, A., Ma, W., Wei, P., Sheng, L., Wang, X., 2024. General debiasing for graph-based collaborative filtering via adversarial graph dropout, in: Proceedings of the ACM Web Conference 2024 (WWW '24), Singapore. doi:10.1145/3589334.3645667, arXiv:2402.13769.

Zhang, A., Sheng, L., Cai, Z., Wang, X., Chua, T.S., 2023. Empowering collaborative filtering with principled adversarial contrastive loss, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023). URL: https://neurips.cc/virtual/2023/poster/71149, arXiv:2310.18700.

Zhang, J., Wang, Y., Song, Q., Hong, L., 2026. ConAlign: Conditional alignment framework for balancing biased and unbiased recommendation. arXiv:2607.24092. preprint.

Zhang, Y., Feng, F., He, X., Wei, T., Song, C., Ling, G., Zhang, Y., 2021. Causal intervention for leveraging popularity bias in recommendation, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 11–20. doi:10.1145/3404835.3462875.

Zhao, M., Wu, L., Liang, Y., Chen, L., Zhang, J., Deng, Q., Wang, K., Shen, X., Lv, T., Wu, R., 2022. Investigating accuracy-novelty performance for graph-based collaborative filtering, in: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 50–59. doi:10.1145/3477495.3532005.

Zhou, H., Chen, H., Dong, J., Zha, D., Zhou, C., Huang, X., 2023. Adaptive popularity debiasing aggregator for graph collaborative filtering, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM. pp. 7–17. doi:10.1145/3539618.3591635.